

Additional Practice

ACC7.4.12

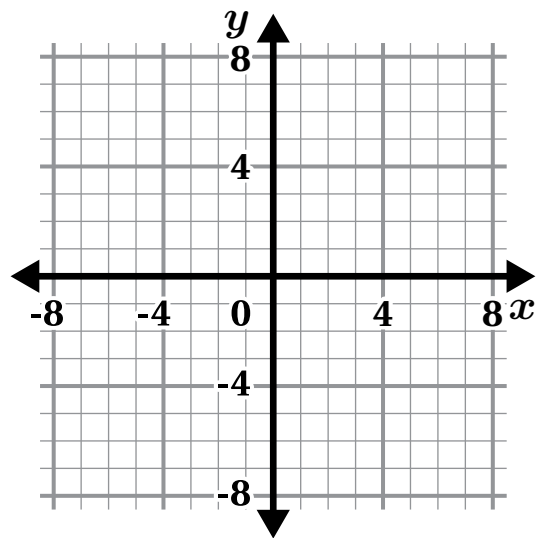
1. Select *all* of the ordered pairs (x, y) that are solutions to the linear equation $3x - 2y = 4$.

- ☐ A. $(0, -2)$ ☐ B. $(2, -1)$
☐ C. $(-2, -5)$ ☐ D. $(-4, 6)$
☐ E. $(4, 4)$

2. The graphs of a linear equation passes through the points $(-1, 2)$ and $(2, 8)$.

Select all the points that are also solutions to this equation. Use the graph if it helps with your thinking.

- ☐ A. $(0, -4)$
☐ B. $(1, 6)$
☐ C. $(-1, 2)$
☐ D. $(-2, 3)$
☐ E. $(-4, -4)$



3. Match each equation with its three solutions.

Equation

Solutions

- | | |
|---------------------------|-----------------------------------|
| a. $2x + y = 5$ | $(9, 3), (5, -1), (8, 2)$ |
| b. $y = \frac{1}{2}x - 1$ | $(3, 2), (0, 0), (-6, -4)$ |
| c. $x - y = 6$ | $(1, 3), (-2, 9), (3, -1)$ |
| d. $2x = 3y$ | $(-4, -1), (4, 1), (16, 4)$ |
| e. $y = \frac{1}{4}x$ | $(2, 0), (4, 1), (-2, -3)$ |

4. Kiran is determining if the ordered pair $(2, -1)$ is a solution to the equation $3x + y = 5$. His work is shown. Is he correct? Explain your thinking.

Kiran's work:

$$3x + y = 5$$

$$3(-1) + 2 = 5$$

$$-3 + 2 = 5$$

$$-1 \neq 5$$

The point $(2, -1)$ is *not* a solution to the equation.

5. Determine whether the following statement is *true* or *false*. Explain your thinking.

The ordered pairs $(10, 38)$, $(-12, -17)$, and $(14, 45)$ all lie on the line whose equation is $y = \frac{5}{2}x + 13$.

Problems 6–7: Complete each table.

6. $y = -\frac{4}{5}x$

x	y
-10	
	4

7. $x + 2y = 7$

x	y
-1	
	2

Additional Practice

ACC7.4.13

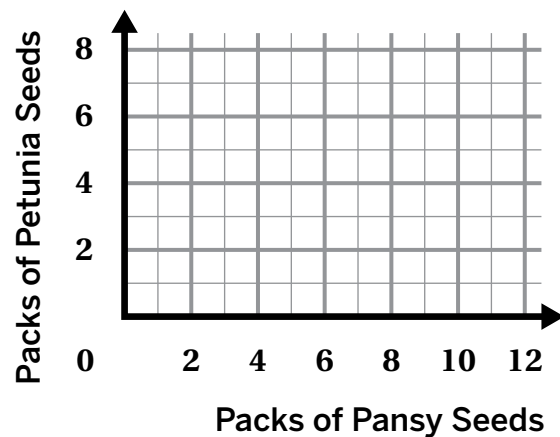
Problems 1–5: Maria has \$48 to spend on flowers for her garden. A pack of pansy seeds costs \$4 each and a pack of petunia seeds costs \$6 each.

- Complete the table to show some possible combinations of packs of pansies and packs of petunias that will cost a total of \$48.

Number of Packs of Pansy Seeds	Number of Packs of Petunia Seeds

- Write an equation that represents the relationship between the number of packs of pansy seeds, x , and the number of packs of petunia seeds, y , that Maria can buy.

- Graph all possible combinations of packs of pansy seeds and packs of petunia seeds that will cost a total of \$48.



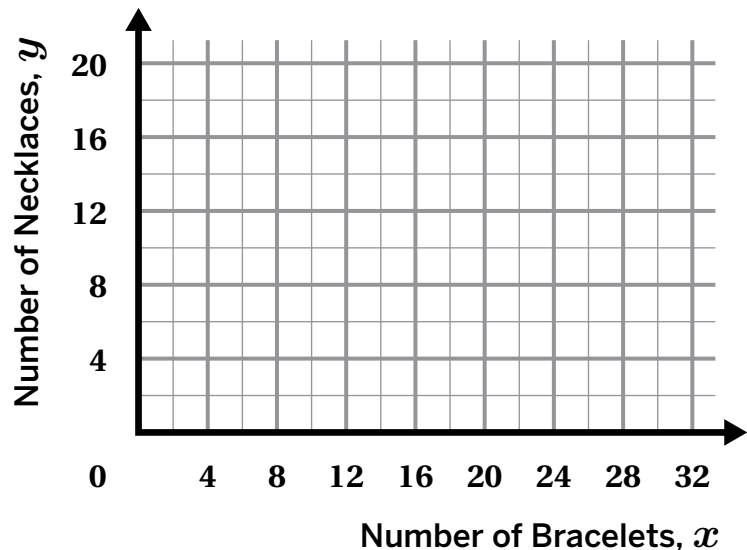
- What is the slope of the line? What does it tell you about this situation?
- What are the x - and y -intercepts of the line? What do they represent in this situation?

Problems 6–8: Jada made bracelets and necklaces to sell at the craft show. She sold her bracelets for \$3 each and her necklaces for \$6 each. She earned \$96 in sales at the craft show.

6. What are two possible combinations of the number of bracelets and necklaces sold that earn a total of \$96 in sales?

7. Write an equation that represents the relationship between the number of bracelets, x , and the number of necklaces, y , that Jada sold.

8. Graph this relationship on the coordinate plane.

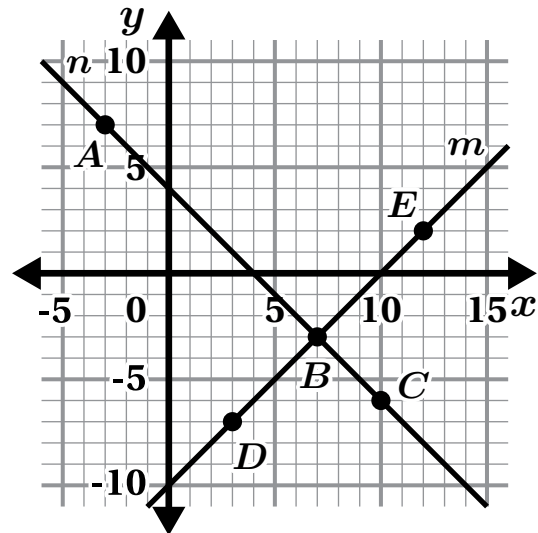


9. What are the x - and y -intercepts of the line? What do they represent in this situation?

Additional Practice

ACC7.4.14

Problems 1–5: Here is a coordinate plane.



1. Determine which line represents this condition:
The coordinates of each point have a sum of 4.

2. Determine which line represents this condition:
The y -coordinate of each point is 10 less than the x -coordinate.

3. Select *all* the points whose coordinates have a sum of 4.

☐ A. Point A

☐ D. Point D

☐ B. Point B

☐ E. Point E

☐ C. Point C

4. Select *all* the points whose y -coordinate is 10 less than the x -coordinate.

☐ A. Point A

☐ D. Point D

☐ B. Point B

☐ E. Point E

☐ C. Point C

5. Select *all* the points whose coordinates have a sum of 4 and whose y -coordinate is 10 less than the x -coordinate.

☐ A. Point A

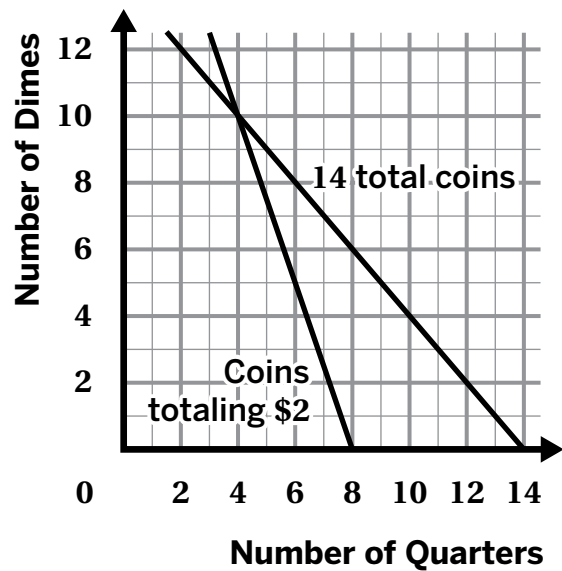
☐ D. Point D

☐ B. Point B

☐ E. Point E

☐ C. Point C

Problems 6–8: On the coordinate plane shown, one line represents combinations of dimes and quarters that have a total value of \$2. The other line represents combinations of dimes and quarters when the total number of coins is 14.



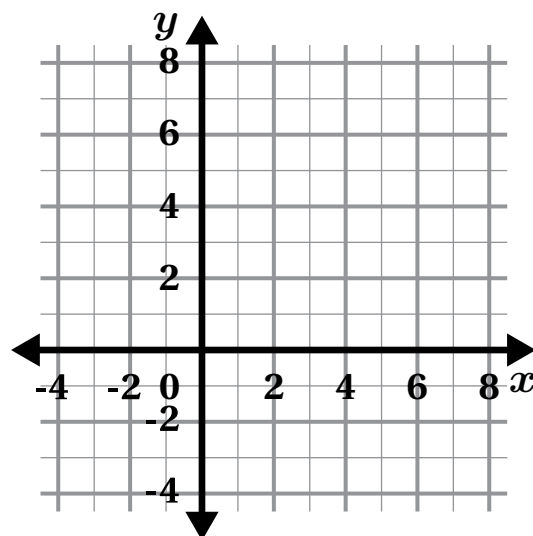
6. Select *all* the combinations of coins that have a total value of \$2.
- ☐ A. 14 quarters and 0 dimes
 - ☐ B. 4 quarters and 10 dimes
 - ☐ C. 6 quarters and 8 dimes
 - ☐ D. 8 quarters and 0 dimes
 - ☐ E. 2 quarters and 12 dimes
7. Select *all* combinations of quarters and dimes that have a total of 14 coins.
- ☐ A. 10 quarters and 4 dimes
 - ☐ B. 8 quarters and 0 dimes
 - ☐ C. 4 quarters and 10 dimes
 - ☐ D. 2 quarters and 12 dimes
 - ☐ E. 4 quarters and 8 dimes
8. What combination of quarters and dimes would equal both 14 coins and total of \$2?
- A. 10 quarters and 4 dimes
 - B. 8 quarters and 0 dimes
 - C. 2 quarters and 12 dimes
 - D. 4 quarters and 10 dimes

Additional Practice

ACC7.4.15

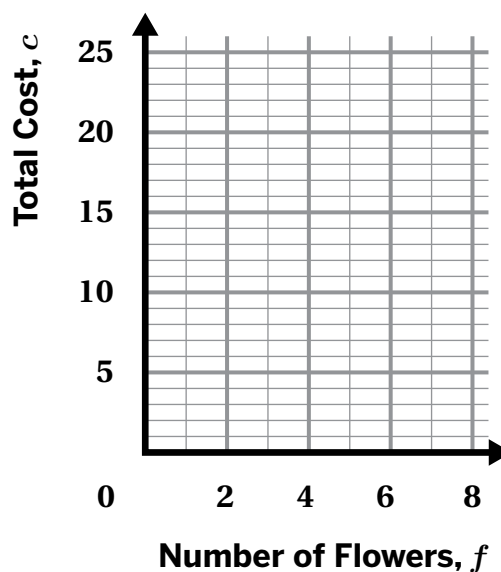
- Two baby elephants, Jack and Yusef, are gaining weight each day. Jack weighs 205 pounds and is gaining 2 pounds per day. Yusef weighs 195 pounds and is gaining 3 pounds per day. Is there a day when they will weigh the same amount? Explain your thinking.

- Draw a graph to determine x - and y -values that make both of the equations $y = 2x + 1$ and $y = -\frac{1}{2}x + 6$ true.



Problems 3–4: Trevor and Angela each bought an arrangement of flowers in a vase. The cost, c , of f flowers for Trevor's arrangement is represented by the equation $c = 4f + 4$. The cost, c , of f flowers in dollars for Angela's arrangement is represented by the equation $c = 2f + 10$.

- Graph the equations for the cost of each arrangement of flowers on the same coordinate plane.
- Identify the point of intersection. What does the intersection point mean in this context?



Problems 5–6: Han and Elena are hosting a school bake sale.

Before the sale, Han received \$5 in donations and will earn \$2.50 per baked good sold.

The table shown represents the amount of money Elena raised, where b represents the number of baked goods sold and t represents the total amount of money raised.

Number of Baked Goods Sold (b)	Total Money Raised (t)
2	14
3	16
4	18

5. How many baked goods must be sold before they raise the same amount of money? Show or explain your thinking.

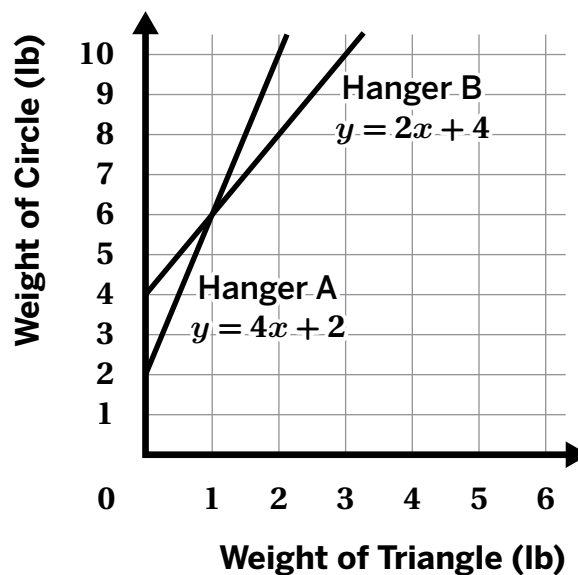
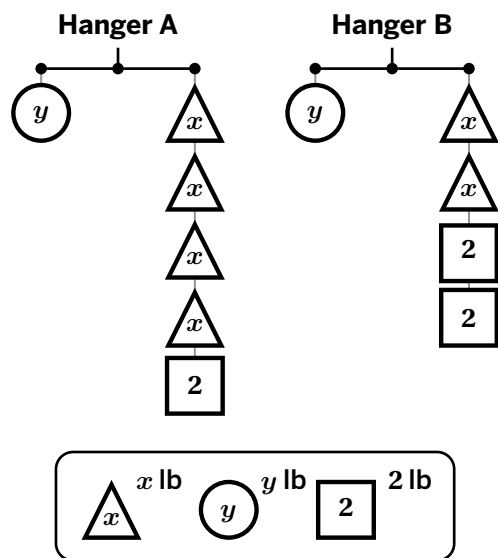
6. How much will each of them have raised at that time? Show or explain your thinking.

7. The point where the graphs of two equations intersect has a y -coordinate of -6 . One equation is $y = 2x - 4$. Determine the other equation if its graph has a slope of -3 . Show or explain your thinking.

Additional Practice

ACC7.4.16

Problems 1–2: The hangers and the graph represent the same system of equations.

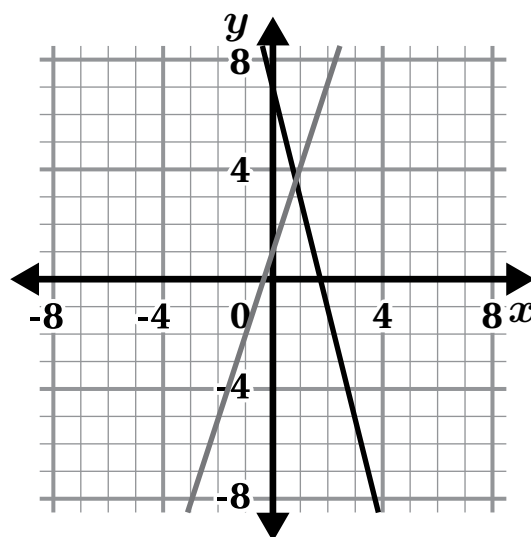


- Determine the solution to the systems of equations.
- What does the solution tell you about the weight of a triangle and the weight of a circle that will balance the hanger?

Problems 3–4: Here is a graph.

3. Write an equation that can represent each line.

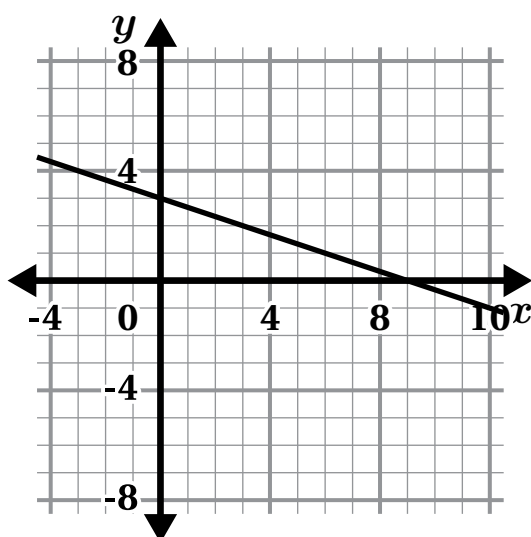
4. Estimate the solution to the system.



Problems 5–7: Here is a graph that represents one equation in a system of equations.

5. Write a second equation for the system so that it has *infinitely many solutions*.

6. Write a second equation whose graph goes through $(0, -2)$ so that the system has *no solution*.



7. Write a second equation whose graph goes through $(0, -4)$ so that the system has one solution $(6, 0)$. Show or explain your thinking.

Additional Practice

ACC7.4.17

1. What is the solution to the system of equations below?

$$y = 6$$

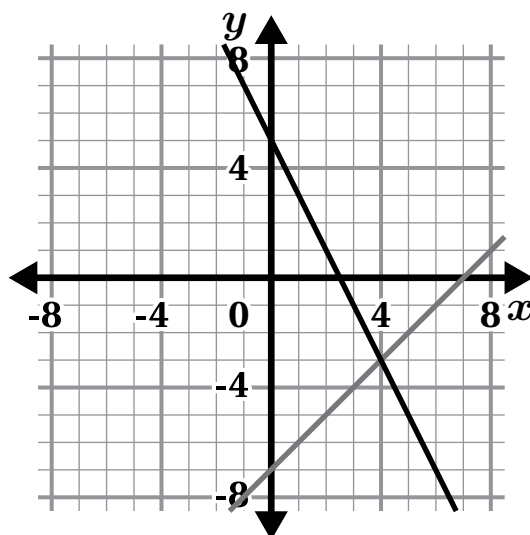
$$y = -\frac{1}{2}x + 4$$

- A. $(6, -4)$
- B. $(-4, -6)$
- C. $(4, 6)$
- D. $(-4, 6)$

Problems 2–3: Here is a graph of the systems of equations:

$$y = x - 7$$

$$y = -2x + 5$$



2. How can you determine the solution to this system of equations by looking at the graph?
3. What is the solution to the system of equations?

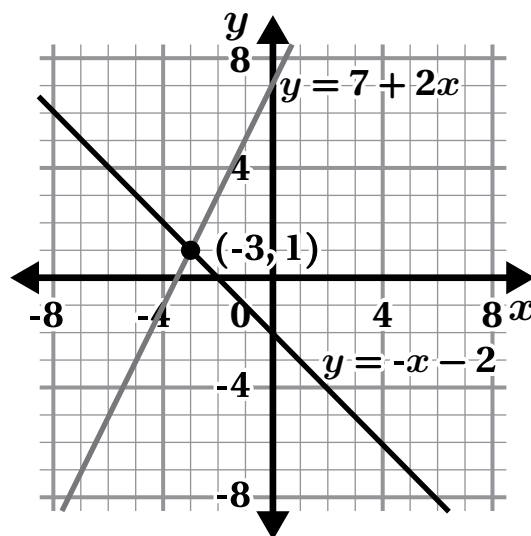
Problems 4–5: Solve each system of equations. Show your thinking.

4. $y = 2x + 5$
 $x = -3$

5. $y = 2x - 4$
 $y = -2x + 12$

Problems 6–8: Use the lines in the graph to decide whether each statement is true or false. Show or explain your thinking.

6. The solution to the equation $7 + 2x = -x - 2$ is $x = 1$.



7. The point $(-3, 1)$ is the solution to the following systems of equations:

$$y = 7 + 2x$$

$$y = -x - 2$$

8. The point $(0, -2)$ is a solution to the equation $y = -x - 2$.

9. The solution to a system of equations is $(-1, -4)$. Select two equations that might make up the system. Show or explain your thinking.

☐ A. $y = 2x + 5$

☐ D. $y = x + 3$

☐ B. $y = x - 3$

☐ E. $y = -\frac{1}{2}x + 1$

☐ C. $y = 3x - 1$

Additional Practice

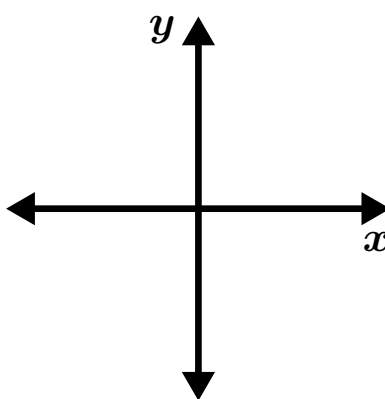
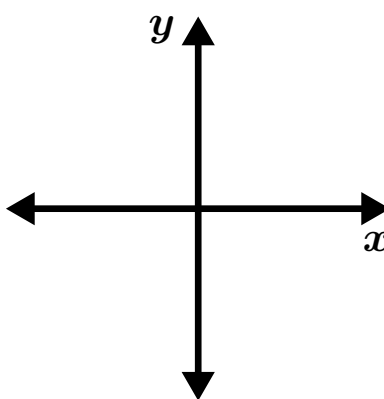
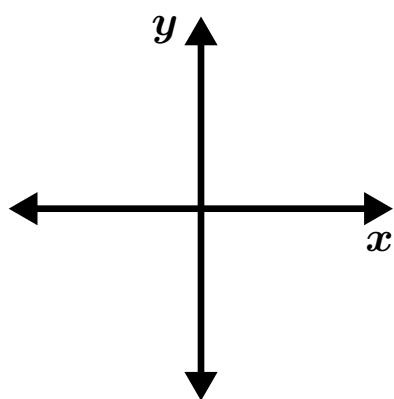
ACC7.4.18

Problems 1–3: Sketch two lines that match each description. Then use the following terms to describe the number of solutions for each system of equations: no solution, one solution, infinitely many solutions

1. Two lines with different slopes but the same y -intercept.

2. Two lines with the same slope and the same y -intercept.

3. Two lines with the same slope and different y -intercepts.



4. Which equation, together with the equation $y = -8x + 4$, creates a system with no solution? Select all that apply.

☐ A. $y = 2(4x + 2)$

☐ D. $y = -8(x - \frac{1}{2})$

☐ B. $y = 8x - 4$

☐ E. $y = 4(-2x - 1)$

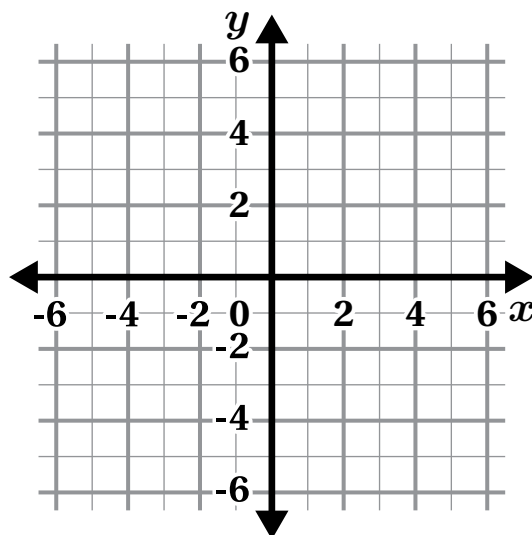
☐ C. $y = -2(4x + 2)$

5. How many solutions does this system have? Use the graph if it helps your thinking.

$$y = \frac{2}{3}x - 4$$

$$y = \frac{1}{3}(2x - 6)$$

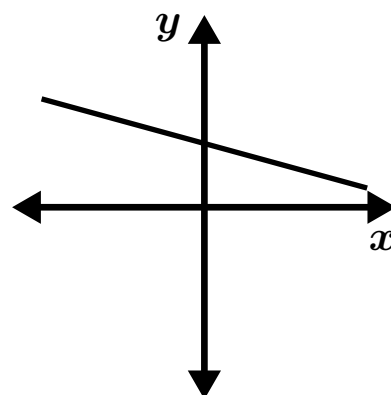
Show or explain your thinking.



Problems 6–7: A graph of a system of equations is provided.

6. Select two equations that could make up the system of equations shown on the graph.

- ☐ A. $y = -\frac{1}{2}x + 4$
- ☐ B. $y = -\frac{1}{2}(x + 4)$
- ☐ C. $y = -\frac{1}{2}(x + 16)$
- ☐ D. $y = -\frac{1}{2}(x - 8)$
- ☐ E. $y = -4 - \frac{1}{2}x$



7. Does this system of equations have no solution, one solution, or infinitely many solutions? Explain your thinking.

Problems 8–9: Alex graphed this system:

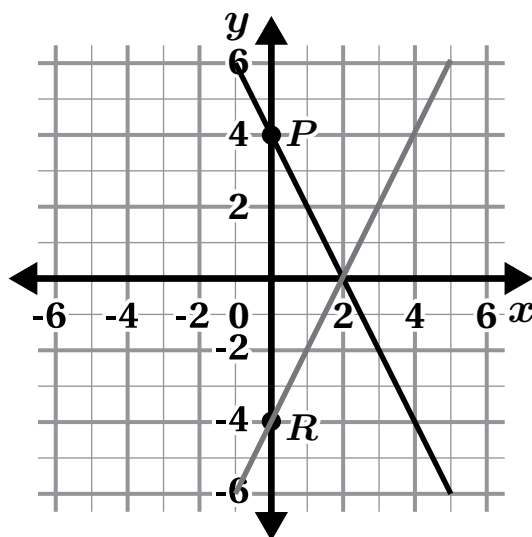
$$y = 2x - 4$$

$$y = -2x + 4$$

He marked its solutions with points P and R .

8. Which statement describes Alex's solutions?

- A. His solutions are correct.
- B. He marked the y -intercepts instead of the x -intercepts.
- C. He marked only the y -intercepts instead of the x - and y -intercepts.
- D. He marked the y -intercepts instead of the intersection point of the two lines.



9. What is the solution to the system of equations?

Additional Practice**ACC7.4.19****Problems 1–6:** Solve each system of equations. Show your thinking.

1. $y = -2x$
 $3x + y = 10$

2. $y = 2x$
 $x = 4y - 14$

3. $y = 6x + 10$
 $y = -2x - 14$

4. $y = -3x + 12$
 $y = 5x - 8$

5. $x = -3$
 $y = 4x + 8$

6. $y = 12$
 $y = \frac{1}{2}x - 4$

Problems 7–9: Here is an incomplete system of equations.

$$y = \frac{4}{5}x + 6$$

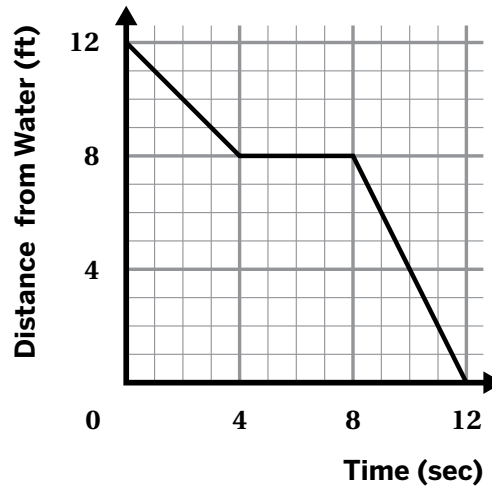
$$y = ???$$

7. Create a second equation so that the system has no solution.
8. Create a second equation so that the system has an infinite number of solutions.
9. Create a second equation so that the system has a solution of $(10, 14)$.

Additional Practice

ACC7.5.01

Problems 1–3: This graph shows a turtle's journey towards the ocean. Determine whether each statement is true or false. If the statement is false, explain your thinking.

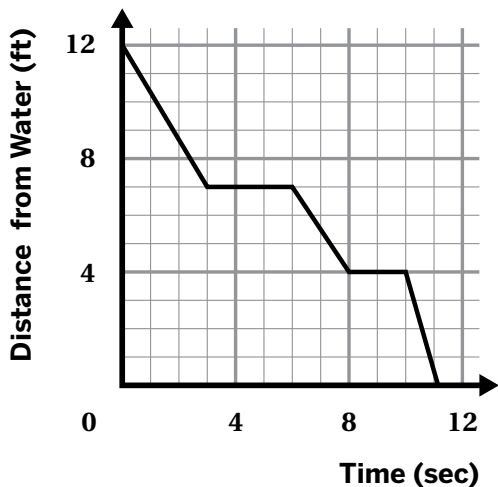


1. The turtle was 4 feet from the water at 4 seconds.

2. The turtle was 12 feet from the water at 0 seconds.

3. The turtle's distance from the water did not change from 4 to 9 seconds.

Problems 4–6: This graph represents another turtle walking across the sand towards the water.



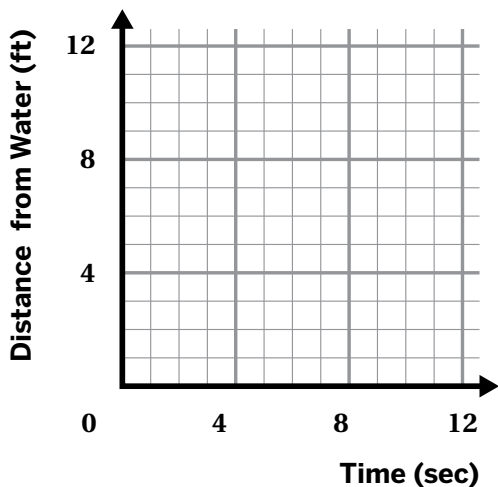
4. What story does the graph tell about the turtle's journey?

5. How far was the turtle from the water after 5 seconds?

6. After how many seconds is the turtle's distance 4 feet from the water?
A. 3 seconds B. 6 seconds C. 8 seconds D. 11 seconds

7. A turtle started their journey towards the ocean. When the turtle started their journey, they were 6 feet from the water. The turtle traveled for 4 seconds. When the turtle was 4 feet from the water, they rested for 4 seconds. Then, the turtle kept traveling towards the water. They reached the water at 12 seconds.

Sketch a graph that could represent the turtle's distance from the water.



Additional Practice

ACC7.5.02

1. Determine whether each table could represent a function. Explain your thinking.

a

Input	Output
4	6
5	7
6	8
4	9
7	10

b

Input	Output
-4	17
-2	5
0	0
2	5
3	17

2. A birthstone is a gemstone that represents the month in which you were born. Both tables show a relationship between birthday and birthstone.

Table A

Birthdate	Gemstone
October 16	Opal
July 5	Ruby
March 20	Aquamarine
June 22	Pearl
October 26	Opal

Table B

Gemstone	Birthdate
Opal	October 16
Ruby	July 5
Aquamarine	March 20
Pearl	June 22
Opal	October 26

Which table(s) represent a function?

A. Table A

B. Table B

C. Neither

D. Both

3. Kiran earns an hourly wage. Determine whether each statement is *true* or *false*. Explain your thinking.

- a Kiran's earnings are a function of his hours worked.
- b Kiran's hours worked are a function of his earnings.

4. A partially completed input-output table is shown. Complete the table so that it represents a function.

Input	Output
−5	25
	9
−1	1
10	

5. Diego wants to know if it is possible for a function to have more than one input value, but only one output value. What would you tell Diego? Use the table if it helps your thinking.

Input	Output

6. Identify a possible rule for the table shown.

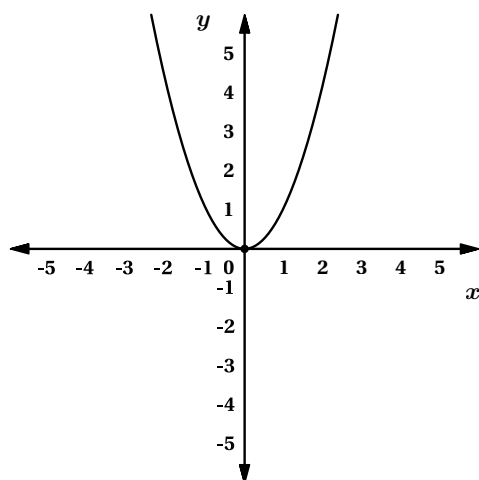
Input	Output
1	1
2	1
0	1
−1	0
−2	0

Additional Practice

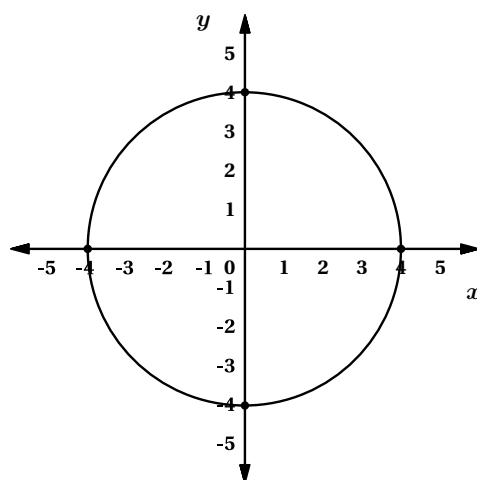
ACC7.5.03

1. Which graph represents y as a function of x ?

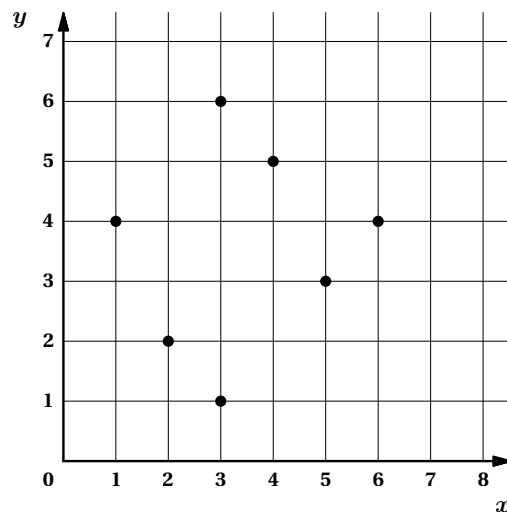
A.



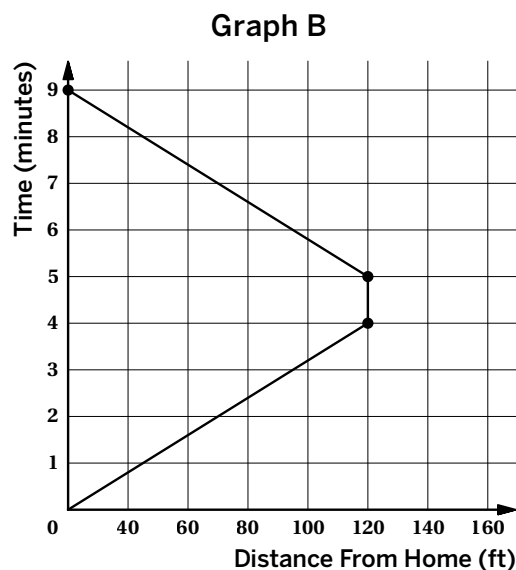
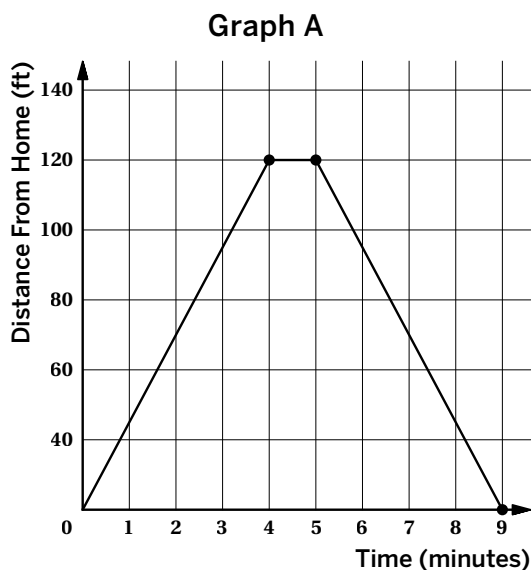
B.



2. Which point on the graph could you remove so that y is a function of x ?



3. Clare rode her bike to her friend's house and back. Both graphs show the relationship between her time and her distance from home.

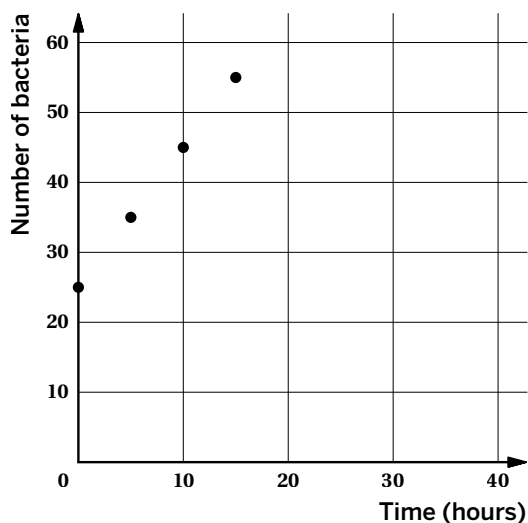


Which graph represents a function?

- A. Graph A B. Graph B C. Both graphs D. Neither graph

Explain your thinking.

4. A scientist graphs the growth of a strain of bacteria. The graph represents a function. Suppose there was the same number of bacteria at 20 and 22 hours. Would the graph still be a function? Explain your thinking.



Additional Practice

ACC7.5.04

1. Which equation expresses the output as a function of the input for the following scenario?

a The amount your school charges, c , for t tickets to the school play that cost \$4 each.

A. $c = \frac{1}{4}t$

B. $t = \frac{1}{4}c$

C. $c = 4t$

D. $t = 4c$

b Identify the independent variable and dependent variable for you answer in part a.

a amount your school charges, c

b tickets, t

Problems 2–4: For each description, write an equation that expresses the output as a function of the input. Then determine the independent and dependent variables.

2. The circumference, C , of a circle with diameter, d .

Equation:

Independent variable:

Dependent variable:

3. The selling price p , after a markup of 15% is applied to the original price of an item, r .

Equation:

Independent variable:

Dependent variable:

4. The area, A , of a square with a side length, s .

Equation:

Independent variable:

Dependent variable:

5. Han's family car averages 25 miles per gallon when driven. Han writes the equation $y = 25x$ to represent the distance, in miles, his family has traveled on a certain number of gallons of gasoline.
 - a What do x and y represent in this situation?
 - b Which variable represents the independent variable? Which variable represents the dependent variable?

6. Mai is buying vegetable plants for her garden. Tomato plants cost \$3 each and pepper plants cost \$1 each. Mai has \$24 to spend on these plants. Let t represent the number of tomato plants Mai buys and p represent the number of pepper plants Mai buys.
 - a Write an equation relating the two variables.
 - b Rewrite the equation so that it expresses p as the dependent variable in terms of t as the independent variable.
 - c Rewrite the equation so that it expresses t as the dependent variable in terms of p as the independent variable.

7. To determine the number of degrees Fahrenheit of a temperature given in degrees Celsius, you multiply the degrees Celsius by $\frac{9}{5}$ and add 32. For each description, write an equation that expresses the output as a function of the input. Then determine the independent and dependent variables.
 - a The temperature in degrees Fahrenheit, F , based on the temperature in degrees Celsius, C .
 - b The temperature in degrees Celsius, C , based on the temperature in degrees Fahrenheit, F .

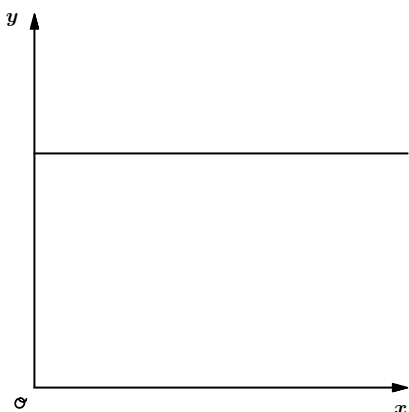
Additional Practice

ACC7.5.05

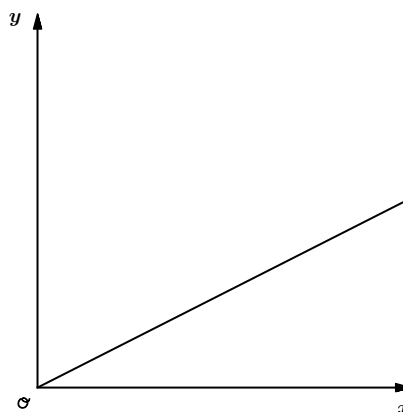
Problems 1–2: Noah walks from his home to the park at a steady pace without stopping.

1. Which graph represents the relationship between Noah's distance from home and time?

Graph A



Graph B



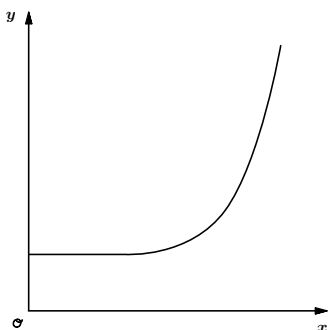
2. Indicate whether the variable described is the possible *independent* or *dependent* variable.

a Distance from home

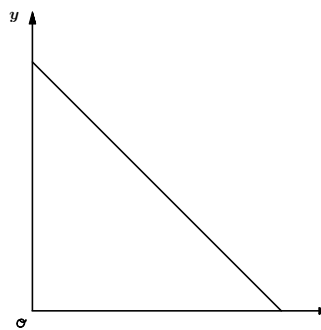
b Time

3. For each scenario, determine which graph best represents it. Then identify the possible independent and dependent variables and how you would label the axes.

Graph 1



Graph 2

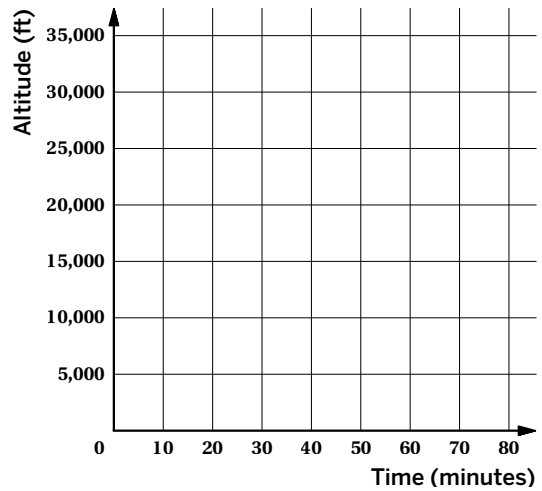


a Mai has some money saved and starts spending the same amount each week.

b The day started cold, but then it got warmer.

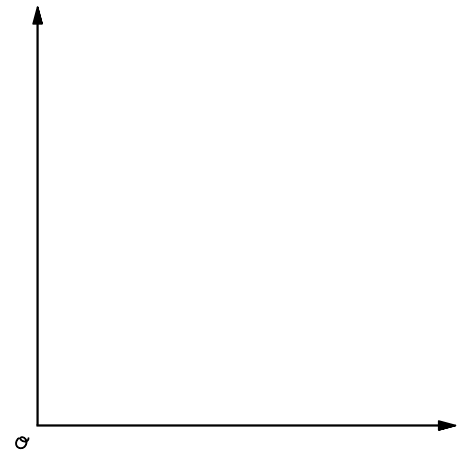
4. An airplane was at an altitude of 5,000 feet. It took the airplane 30 minutes to go from an altitude of 5,000 feet to 30,000 feet. Once at that altitude, the airplane flew for 20 minutes. Then it took the plane 20 minutes to reach an altitude of 10,000 feet.

Sketch a graph that represents this situation.



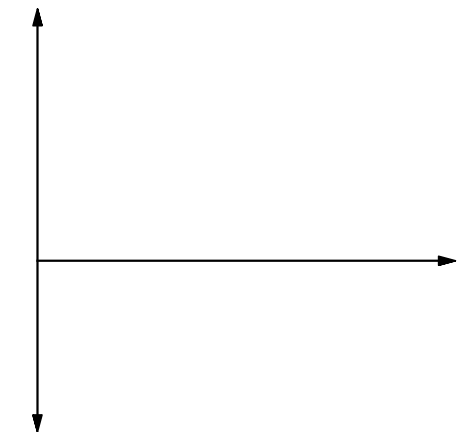
5. Tyler got on his rollerblades. He was increasing his speed but then fell down. He got up and continued rollerblading, gradually increasing his speed.

Sketch a graph that shows Tyler’s speed, in miles per hour, after several minutes. Be sure to label the axes.



6. Kiran was on a water slide. He slid down the slide and dove into the water. He was below the surface water and then swam back to the surface of the water. He swam at the surface of the water for a few minutes.

Sketch a graph that shows Kiran's distance from the surface of the water as a function of time in minutes. Be sure to label the axes.

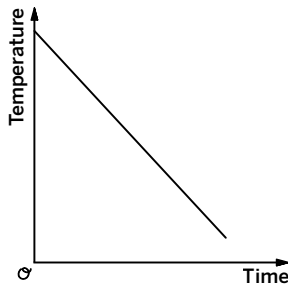


Additional Practice

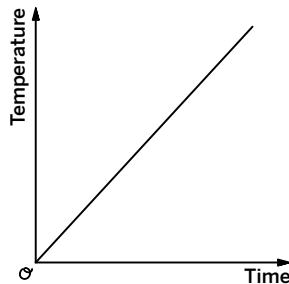
ACC7.5.06

1. Which graph(s) represents an increasing function? Select *all* that apply.

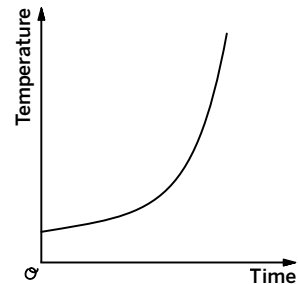
☐ A



☐ B

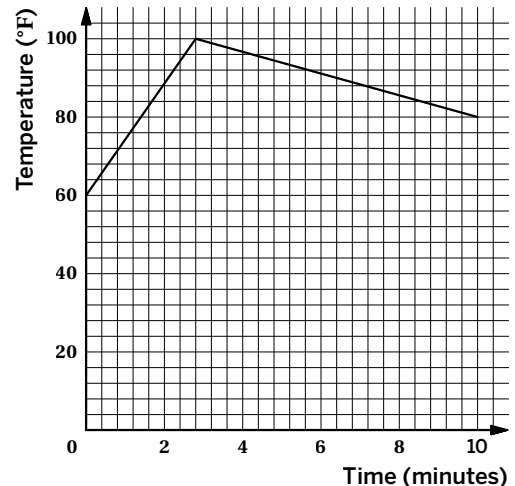


☐ C



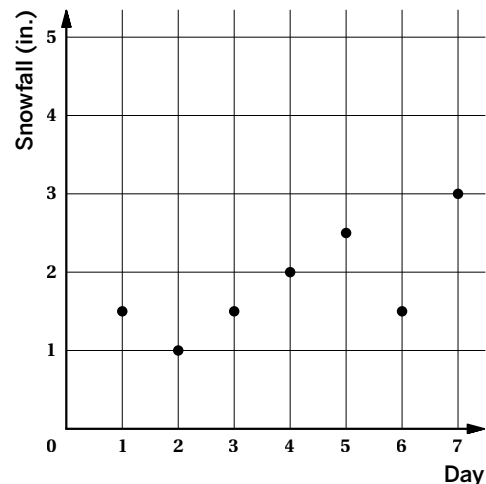
2. The graph shows the temperature of a liquid in a mug over time. Which best describes the graph of the function?

- A. The temperature of the liquid slowly increases, and then slowly decreases over time.
- B. The temperature of the liquid quickly decreases, and then slowly increases over time.
- C. The temperature of the liquid quickly increases, and then decreases even faster over time.
- D. The temperature of the liquid quickly increases, and then decreases more slowly over time.

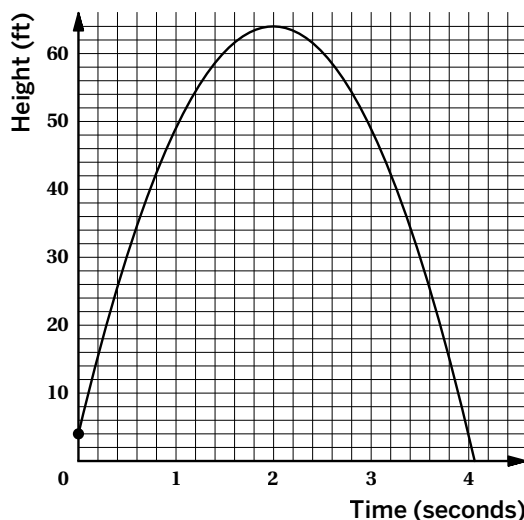


3. The graph shows the total amount of snow that fell each day over a 7-day period.

- a What was the amount of snowfall for Day 2?
- b On which day(s) was the amount of snowfall 1.5 in.?
- c Is the amount of snowfall a function of the day or is the day a function of the amount of snowfall? Explain your thinking. Then determine the independent and dependent variables.

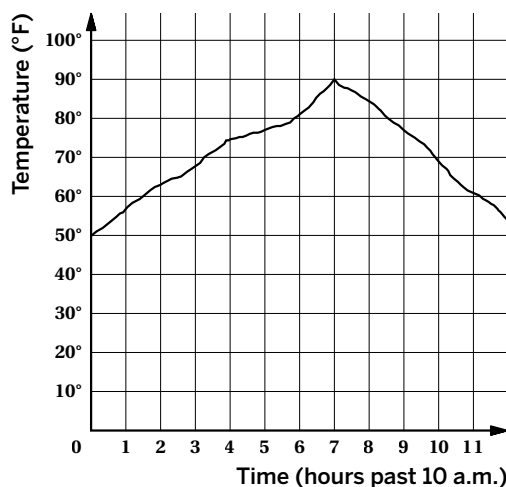


- 4.** Andre's science class is launching model rockets from a picnic table. The graph represents a model rocket that is launched upward, that then falls to the ground.



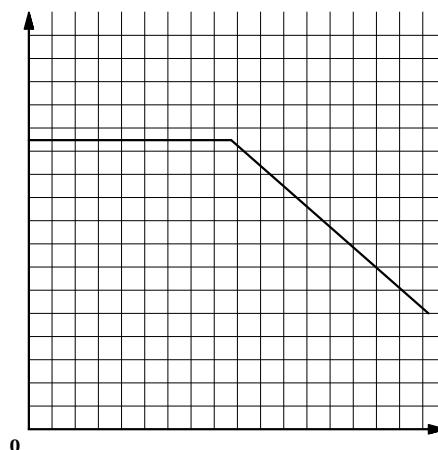
- a** What is the height at which the model rocket was launched?
- b** Plot the point that represents the greatest height of the model rocket. How long did it take for the model rocket to reach that height?
- c** Determine one time interval when the height of the object was increasing.
- d** Determine one time interval when the height of the object was decreasing.

- 5.** The graph shows the temperature between 10 a.m. and 10 p.m. for one day a city.



- a** When the input is 7, what is the output? What does that tell you about the time and temperature?
- b** Determine one time interval when the temperature was increasing.
- c** Determine one time interval when the temperature was decreasing.

- 6.** Describe a real-world situation that would represent the graph. Label the axes.



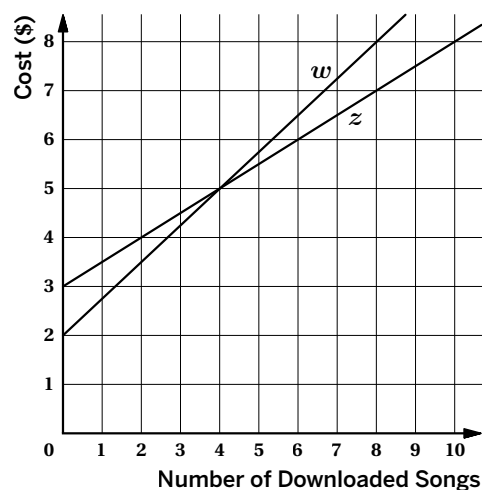
Additional Practice

ACC7.5.07

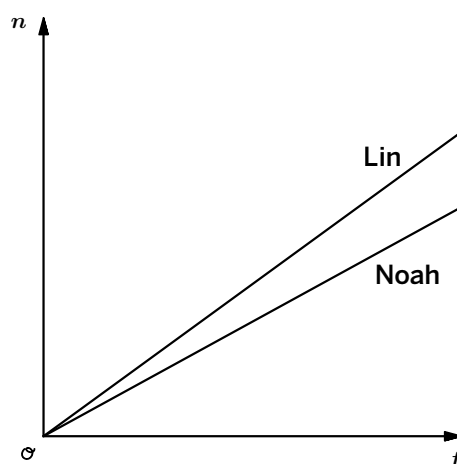
1. Andre is considering two online music subscriptions. Company A charges a flat fee of \$3 plus \$0.50 per downloaded song. Company B charges a flat fee of \$2 plus \$0.80 per downloaded song. Match the companies to the lines w and z shown on the graph.

Company A is represented by line _____

Company B is represented by line _____



2. Noah and Lin are having a typing contest. The graph shows the number of words typed, n , for each student from the start of the contest as a function of time, t . Who is typing faster? Explain your thinking.

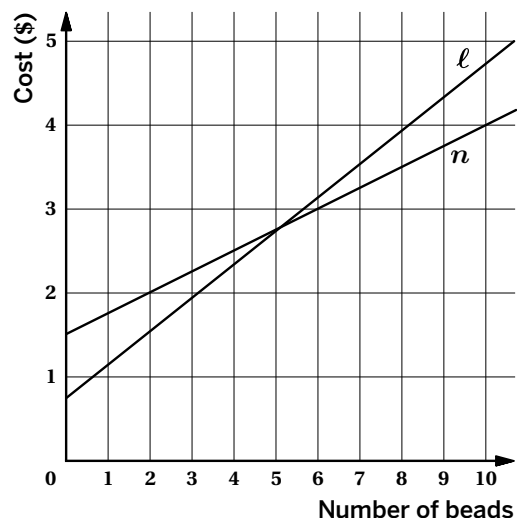


3. Mai and her sibling each want to buy a new scooter. They plan to borrow money from their parents. They each plan to repay their parents the same amount of money every week, but Mai's scooter costs less than her sibling's scooter. Suppose there is a graph that shows the amount they owe their parents, in dollars, from when they begin paying their parents back.

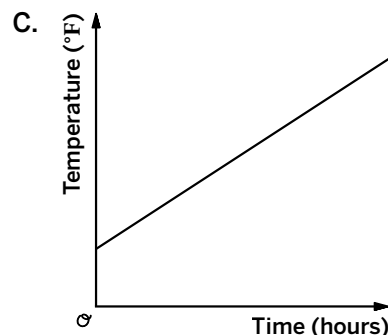
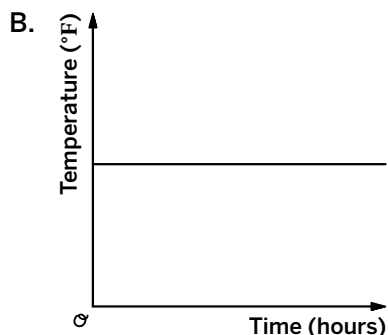
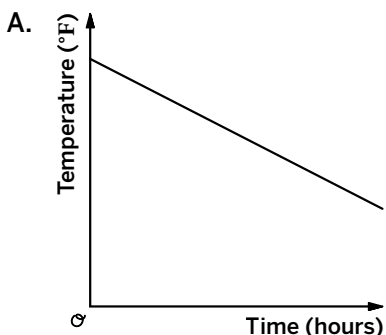
- As you read the graphs from left to right, would you expect the lines to increase or decrease?
- How would you expect the lines representing Mai and her sibling to be different? Explain your thinking.
- How would you expect the lines representing Mai and her sibling to be alike? Explain your thinking.

4. Shawn and Elena each sell customized bracelets. Shawn charges \$0.75 plus \$0.40 per bead added to the bracelet. Elena charges \$1.50 plus d dollars per bead added to the bracelet.

- a Match each person to the lines ℓ and n shown on the graph.
- b For Elena, is the additional charge per bead greater than or less than \$0.40? Explain your thinking.



5. One day, a certain city's temperature steadily increased from noon to 6 p.m. Then from 6 p.m. until midnight its temperature steadily decreased.



- a Which of the graphs is most likely to represent the temperature from noon to 6 p.m.? Explain your thinking.
- b Which of the graphs is most likely to represent the temperature from 6 p.m. to midnight? Explain your thinking.
- c Why does the other graph not likely represent the temperature during either time span? Explain your thinking.

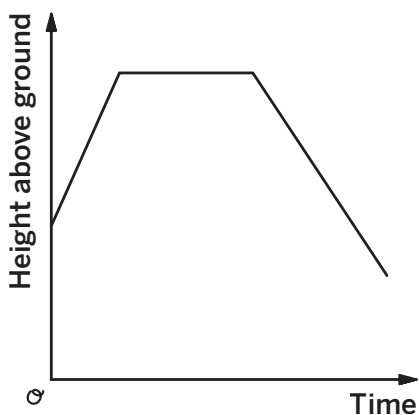
6. Fitness Center A charges a flat fee of \$25 plus \$2 per visit. The cost of Fitness Center B is represented by $y = 2x + 20$ where x is the number of visits. Which fitness center would you choose? Explain your thinking.

Additional Practice

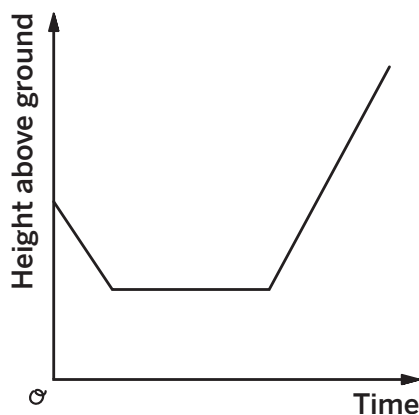
5.09

1. The graph shows the height above ground of two elevators over time. Match each graph to the situation it represents.

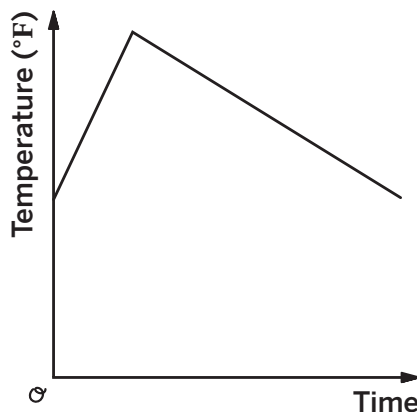
Graph 1



Graph 2

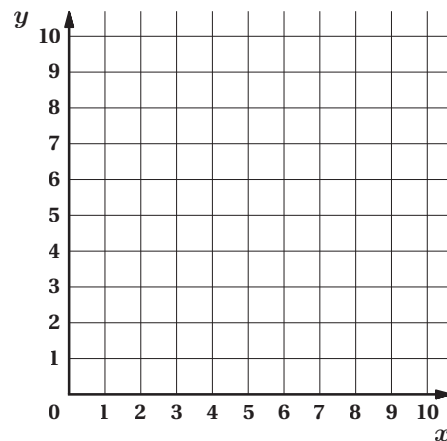


- a The elevator starts at the fourth floor, then stops at the second floor, and then goes up to the seventh floor.
- b The elevator starts at the third floor, then stops at the sixth floor, and then goes down to the second floor.
2. Mai had a teacup of water. The graph shows the temperature of water in a teacup. Describe a possible situation that could represent the graph.



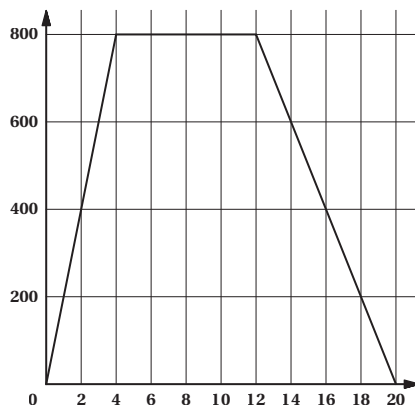
3. The following statements are descriptions of the linear segments that make up a piecewise function. Draw a graph of the piecewise function described.

- Starts at the y -intercept 1.
- Increasing at its greatest constant rate from $x = 0$ to $x = 3$.
- Has a slope of 0 from $x = 3$ to $x = 6$.
- Decreasing at a constant rate from $x = 6$ to $x = 8$.
- Has a slope of 0 from $x = 8$ to $x = 10$.



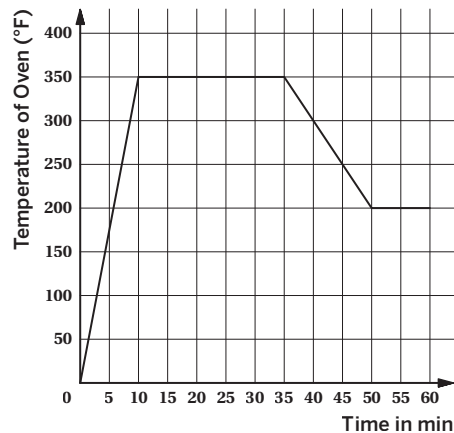
4. Clare walked her dog to the dog park and let the dog play at the park. Then she walked home.

- a The graph shows her distance walked in yards, as a function of time, in minutes. Add the axes labels to the graph to show this.
- b When did she reach the dog park?
- c How far did Clare walk to get to the dog park?
- d How long were Clare and her dog at the dog park?
- e At what rate did she walk to the dog park?
- f At what rate did she walk home from the dog park?



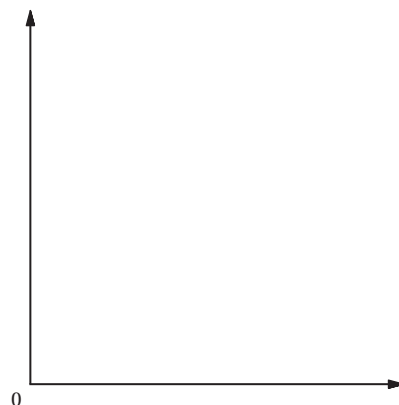
5. Andre's dad is baking a casserole. He preheated the oven. The graph shows the temperature of the oven from when he turned the oven on.

- a When was the temperature of the oven increasing?
- b What was happening from 10 minutes to 35 minutes after the stove was turned on?
- c What rate did the temperature of the oven increase?



6. Noah graphed the growth of his flower. The flower grew at a steady rate until it reached its full height and then stopped growing.

- a What does Noah's graph look like?
- b Han suggested that Noah's flower had a constant growth rate during this time because all parts of the graph can be represented by straight lines. Is he correct? Explain your thinking.

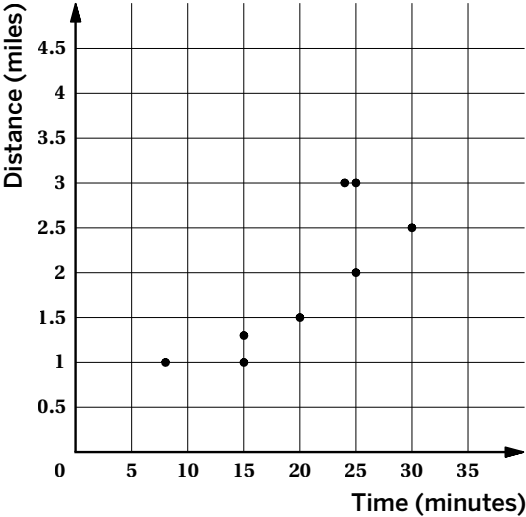


Additional Practice

ACC7.6.01

1. The graph shows the relationship between the amount of time, in minutes, and the distance, in miles, ran by eight different runners. Determine the coordinates for each point on the scatter plot.

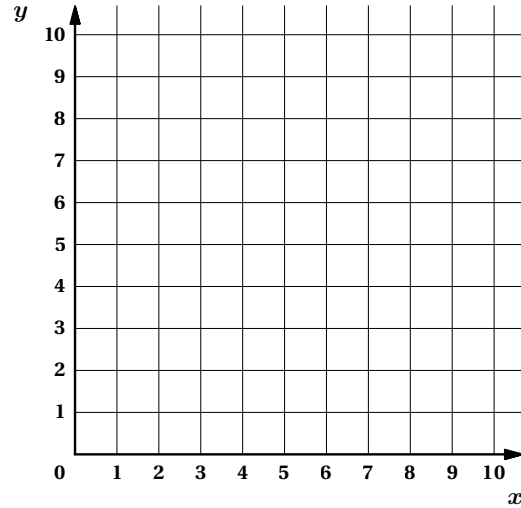
Time (minutes), <i>x</i>	Distance (miles), <i>y</i>



2. Refer to the scatter plot in Problem 1. What patterns do you notice in the data? Explain your thinking.

3. Create a scatter plot from the data shown in the table.

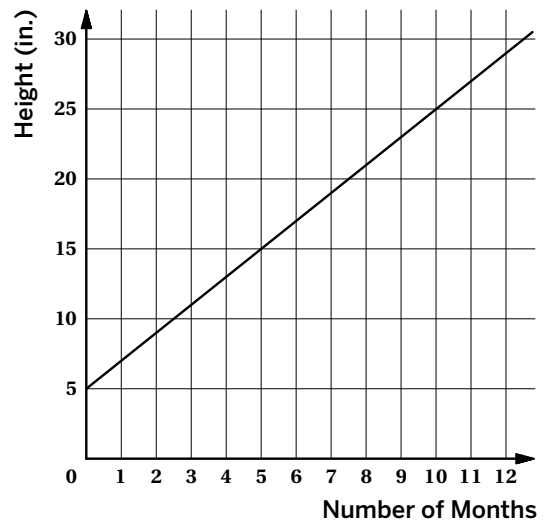
x	y
3	2
4	4
2	5
7	5
4	3
3	6
1	8
6	6



4. When is it helpful to use a table to display data instead of a scatter plot?
Explain your thinking.

5. The graph shows the height, h in inches, of a house plant m months after it has been planted.

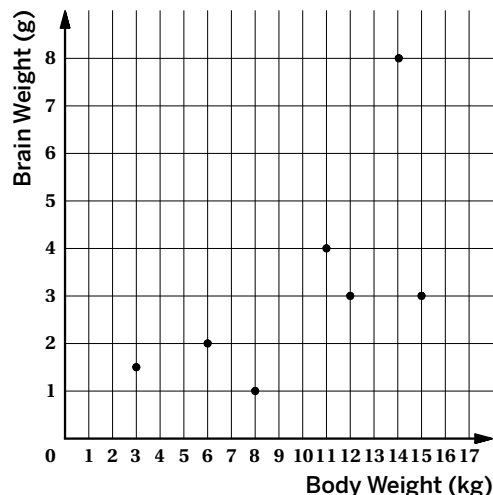
- a Write an equation that gives the plant's height h after m months.
- b After how many months will the plant be 95 inches tall? Show or explain your thinking.



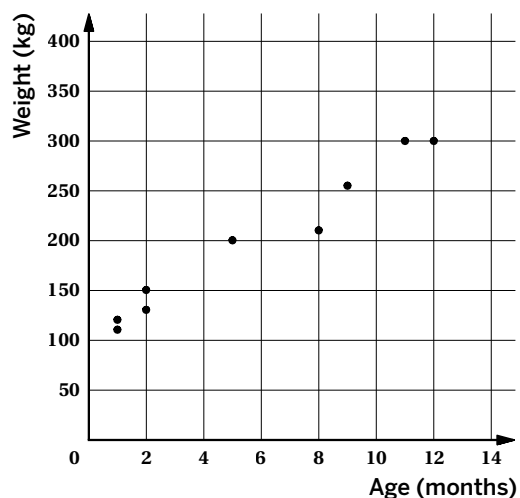
Additional Practice

ACC7.6.02

- The scatter plot shows the body weight, in kilograms, and brain weight, in grams, of seven small animals. Circle the point on the graph that represents the animal with the lightest brain.

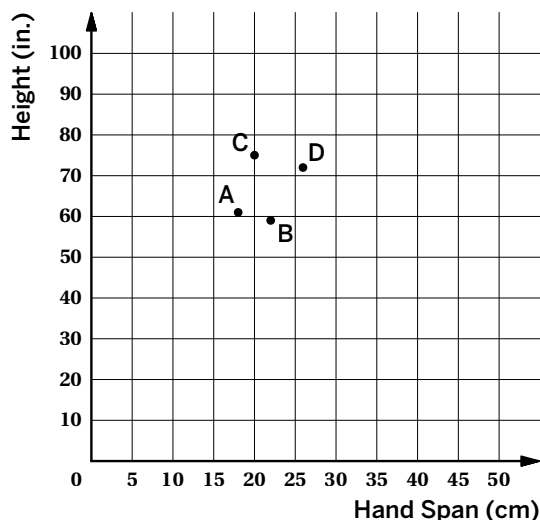


- The scatter plot shows the age, in months, and weight, in kilograms, of nine elephants. Which elephant(s) weigh the most? Circle the point(s) on the graph. Then list the age(s) and weight(s) of the elephant(s).

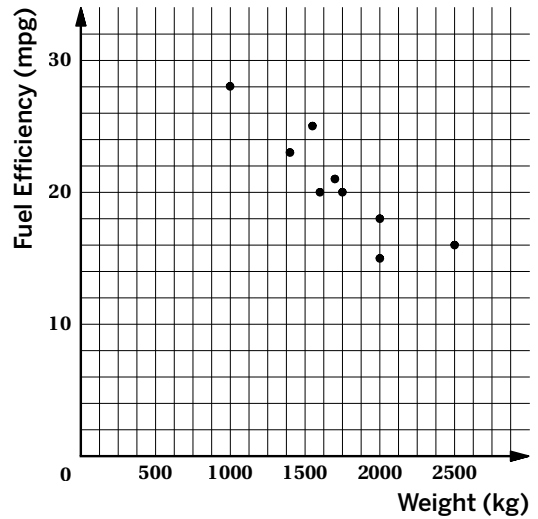


- The table shows the hand span, in centimeters, and the height, in inches, of five middle school students. Plot the point for Student E on the graph.

Student	Hand Span (cm)	Height (in.)
A	18	61
B	22	59
C	20	75
D	26	72
E	20	72



4. The fuel efficiency of a vehicle can be expressed as the distance traveled per unit of fuel used, such as miles per gallon (mpg). The scatter plot shows the weight and fuel efficiency of nine different cars.

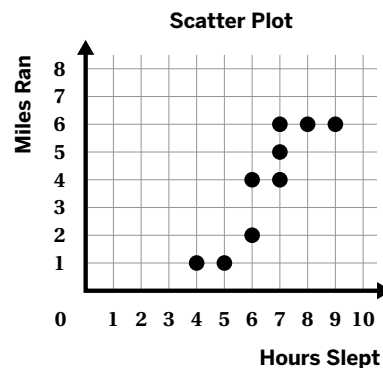
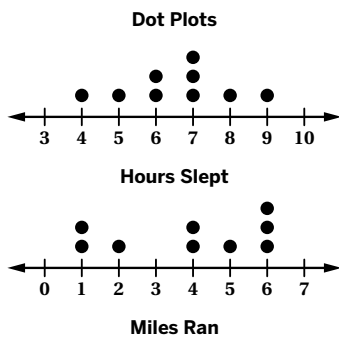


- a How much does the heaviest car weigh?
- b What is the weight and fuel efficiency of the car with the highest fuel efficiency?
- c Plot a point on the graph that represents a car that has a fuel efficiency less than 20 miles per gallon and a weight of 1,500 kilograms.

5. The table from a local clothing store compares the average monthly temperature, in degrees Celsius, to coat sales, in dollars. Elena plans to create a graph to represent this data. What does the point (13, 840) represent?

Temperature (degrees Celsius)	Coat Sales (dollars)
1	1,245
3	1,040
7	1,115
13	840
24	235

6. Students collected data about how many miles athletes ran and how many hours they slept the night before a race. Their data is shown.



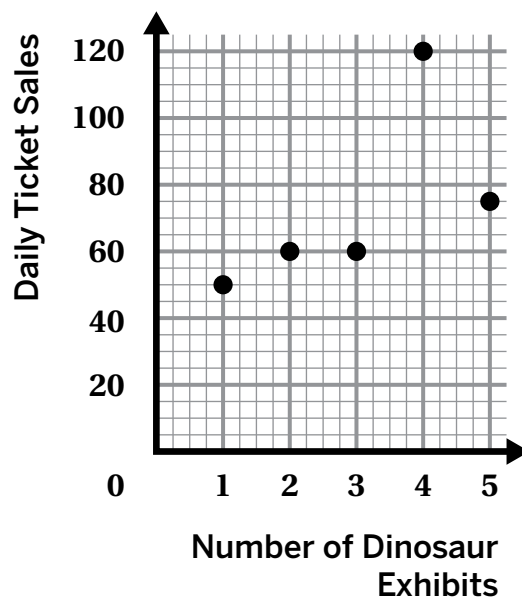
What is different about the two ways of representing the data?

Additional Practice

ACC7.6.03

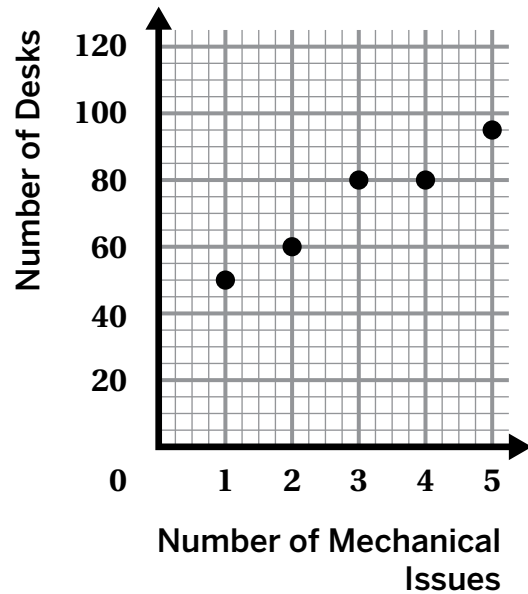
Problems 1–2: A study gathered data about different natural history museums. The table and scatter plot show the number of dinosaur exhibits and the number of daily ticket sales for each museum.

Museum	Dinosaur Exhibits	Daily Ticket Sales
National Dinosaur Museum	4	120
Time Capsule Institute	2	60
The Hall of History	1	50
The Museum of National Treasures	5	75
The Hall of Historical Tales	3	60



- Circle the point on the scatter plot that represents the data for the National Dinosaur Museum.
- What does the point (1, 50) represent?

Problems 3–6: A factory produces wood desks for offices. This scatter plot shows the number of mechanical issues and the number of desks produced for five machines at a factory.



3. What is the number of desks produced at a machine with the most mechanical issues?
4. What is the number of mechanical issues for a machine that produces the least number of desks?
5. The number of desks produced at a machine that has 0 mechanical issues is 60 desks. Plot a point on the graph that represents this machine.
6. What does the point (3, 80) represent?

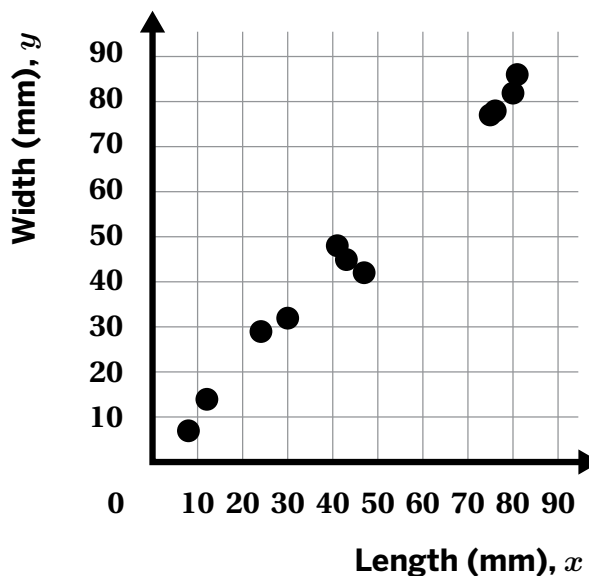
Additional Practice

ACC7.6.04

- This scatter plot shows the length and widths, in millimeters, of several bird eggs.

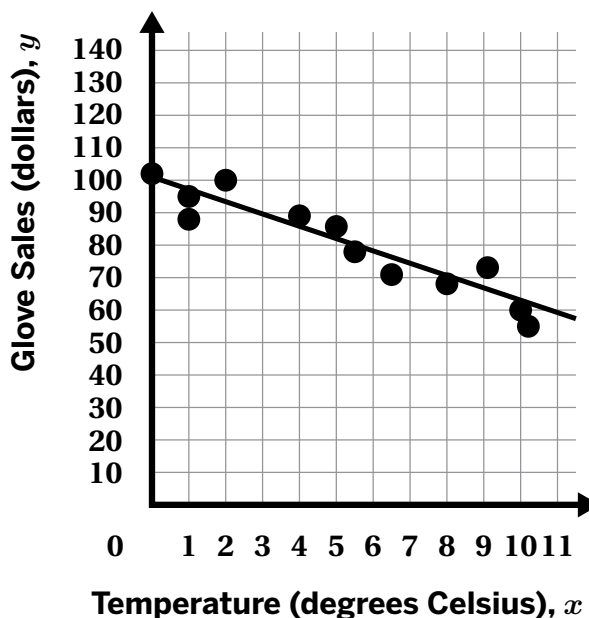
Which is the best prediction for the width of a bird egg if the length is 60 millimeters?

- 31 millimeters
- 42 millimeters
- 50 millimeters
- 62 millimeters



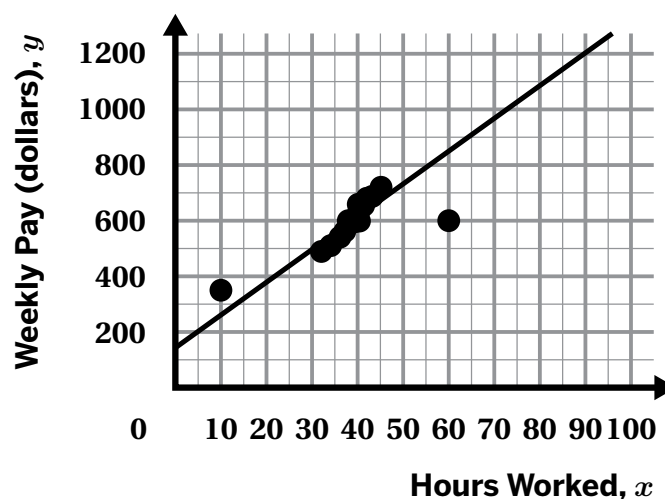
Problems 2–4: This scatter plot shows the average daily temperature, in degrees Celsius, and glove sales, in dollars, at a local store for the past 12 days, along with the linear model $y = -3.8x + 101$.

- What were the glove sales when the temperature was 5.5°C ?
- How much in glove sales does the linear model predict a day with a temperature of 7°C will have?
- If the temperature is 15°C , how much in glove sales does the linear model predict the store will have?



Problems 5–6: This scatter plot shows several workers' weekly hours and pay, along with a linear model represented by the equation $y = 11.8x + 143$.

5. Use the linear model's equation to predict the weekly pay for a person who works 20 hours.

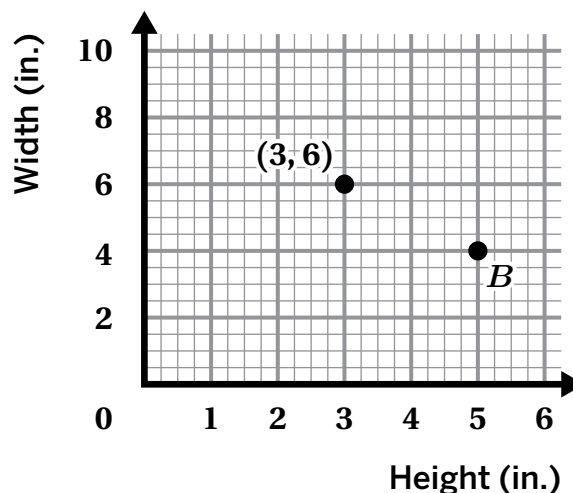


6. Does the scatter plot appear to have any outliers? If so, circle them and describe what they represent about weekly hours worked and pay.

Additional Practice

ACC7.6.05

Problems 1–3: The graph shows the height and width of sketchbooks available at an art store.



1. What does the point (3, 6) tell you about the sketchbook?

2. Plot a point to represent a different sketchbook that has a height of 4 inches and a width of 9 inches.

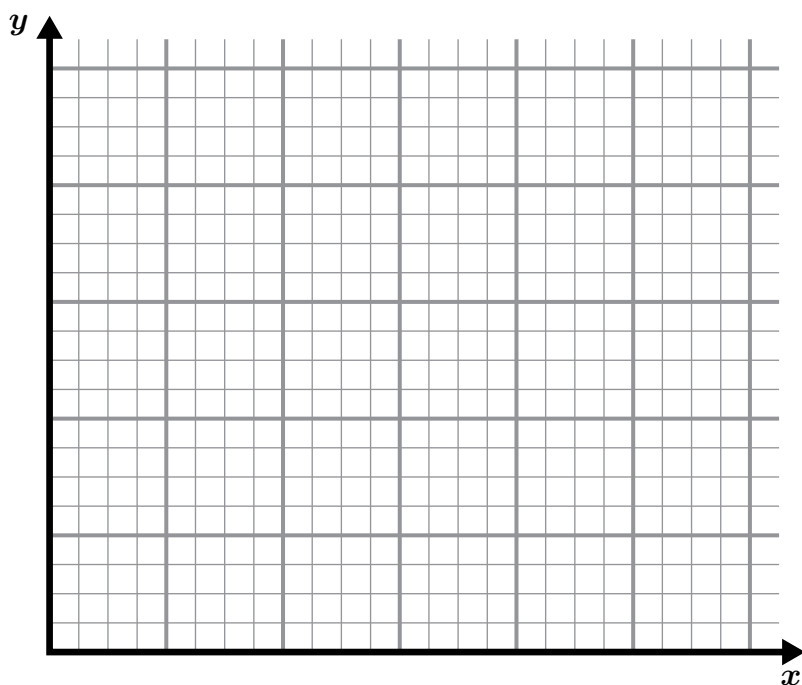
3. What coordinates represent point *B*?

- A. (5, 4)
- B. (4, 5)
- C. (4, 4)
- D. (4, 6)

Problems 4–5: Here is a table.

x	y
1	2
2	5
3.5	6
4	8
5	8.5
5.5	9

- 4.** Create a scale for the graph so it fits all the data. Then create a scatter plot of the data.

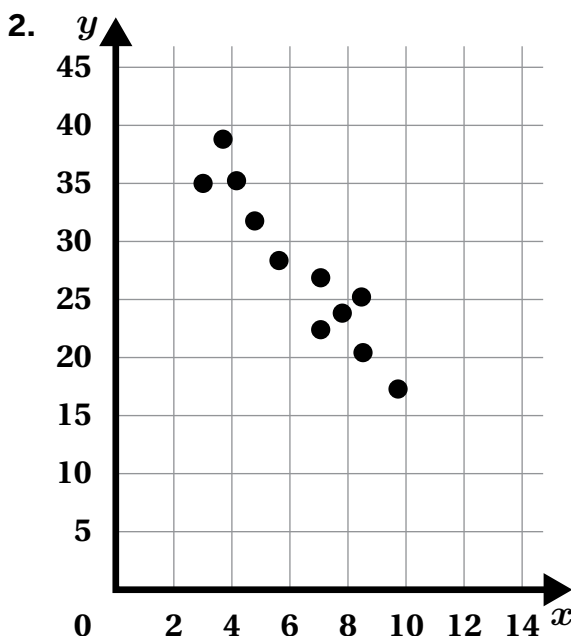
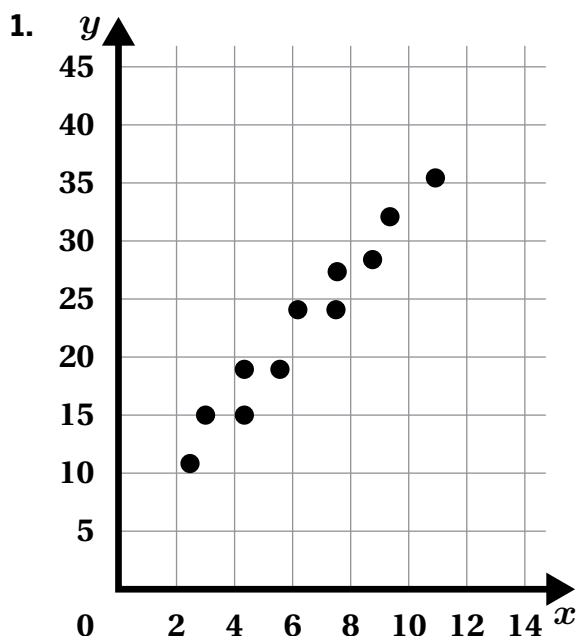


- 5.** What pattern do you notice in the scatter plot?

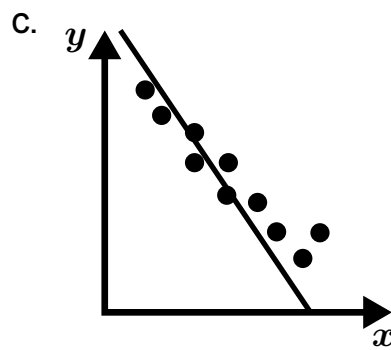
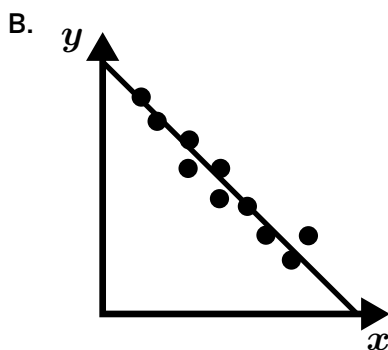
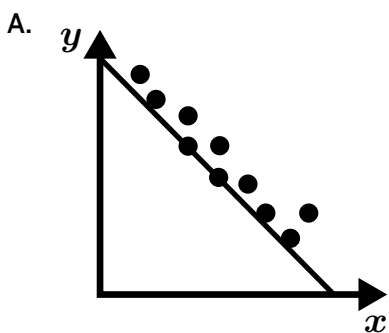
Additional Practice

ACC7.6.06

Problems 1–2: Sketch a line of fit for the data.



3. Which line best fits the data? Circle one.



Explain your thinking.

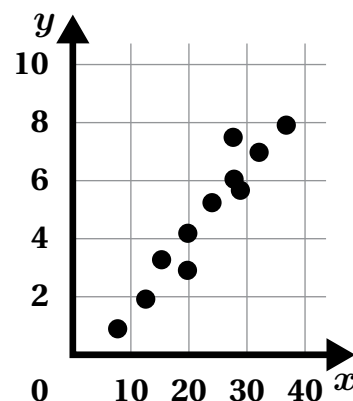
Additional Practice

ACC7.6.07

1. Which type of association does the scatter plot show?

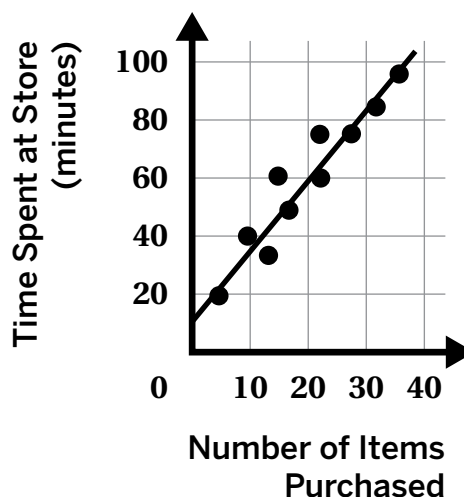
- A. Positive association
- B. Negative association
- C. No association

Explain your thinking.



Problems 2–4: The scatter plot shows the data from 10 customers shopping at a supermarket, along with a linear model whose equation is $y = 2.1x + 15.6$.

2. What is the slope of the linear model?



3. What does the slope represent in this situation?

4. What type of association is there between the number of items purchased and time spent at the store? Circle your choice.

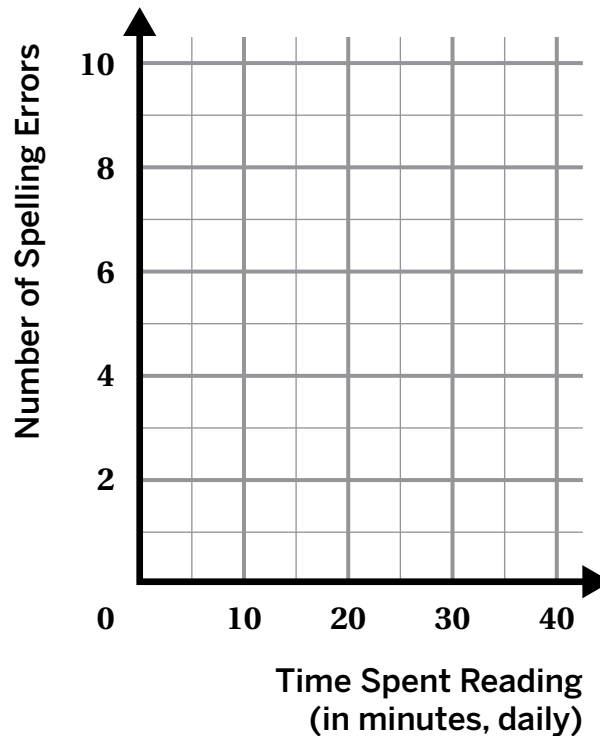
Positive association Negative association No association

Explain your thinking.

Problems 5–6: Librarians looked at the relationship between the minutes spent reading a book (daily) and the number of spelling errors made when writing an email. They found the variables had a negative association.

5. What does this negative association mean about the relationship between time spent reading and the number of spelling errors when writing an email?
 - A. As the time spent reading increases, the number of spelling errors decreases.
 - B. As the time spent reading increases, the number of spelling errors increases.
 - C. As the time spent reading decreases, the number of spelling errors decreases.
 - D. There is no relationship between the time spent reading and the number of spelling errors.

6. Create a scatter plot that represents this situation.

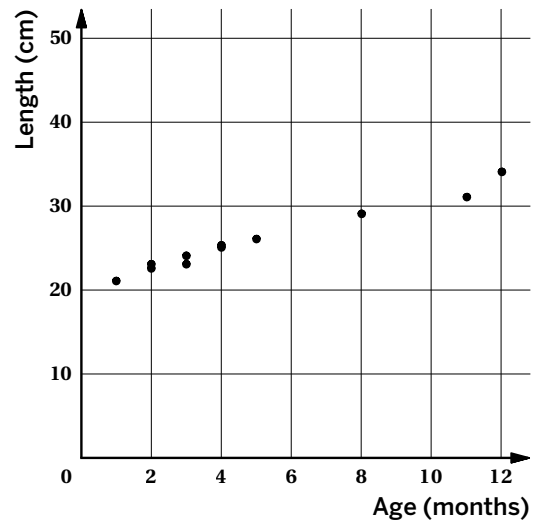


Additional Practice

ACC7.6.08

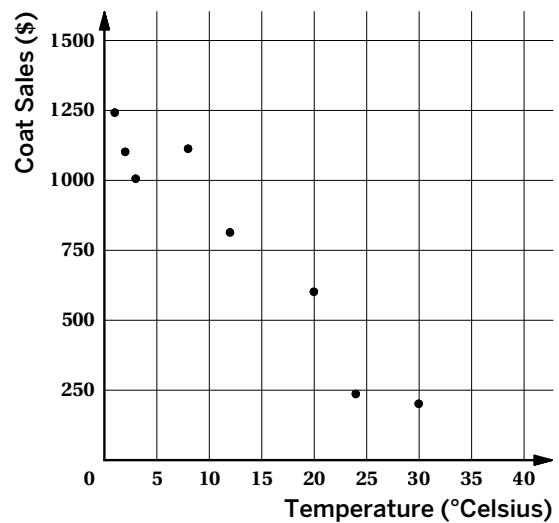
1. Select *all* the terms that describe the association on the scatter plot.

- ☐ A. Linear association
- ☐ B. Non-linear association
- ☐ C. Positive association
- ☐ D. Negative association
- ☐ E. No association

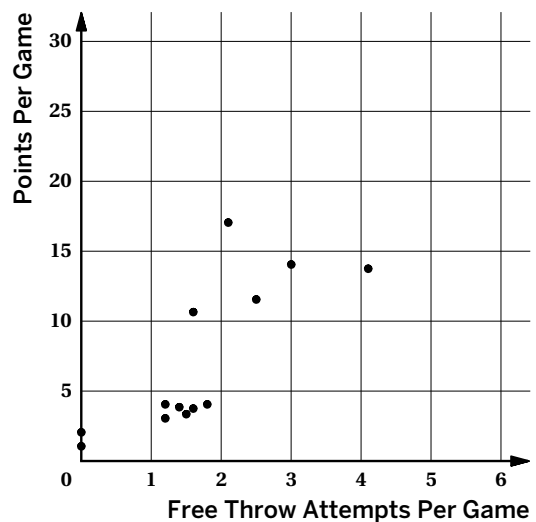


2. Select *all* the terms that describe the association on the scatter plot.

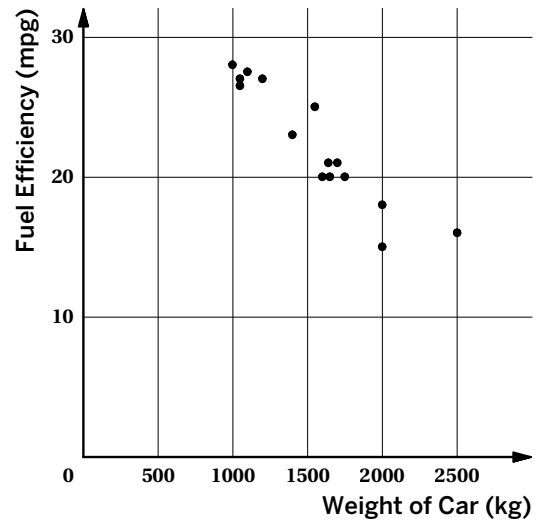
- ☐ A. Linear association
- ☐ B. Non-linear association
- ☐ C. Positive association
- ☐ D. Negative association
- ☐ E. No association



3. Circle any clusters in the data. Use the labels on the axes to explain what the clusters mean in context.

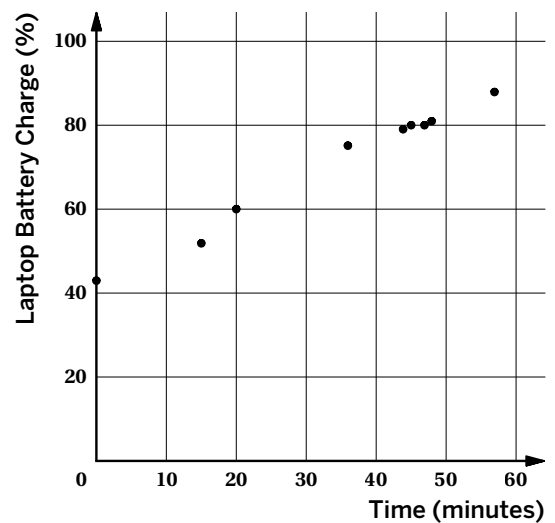


4. Circle any clusters in the data. Use the labels on the axes to explain what the clusters mean in context.

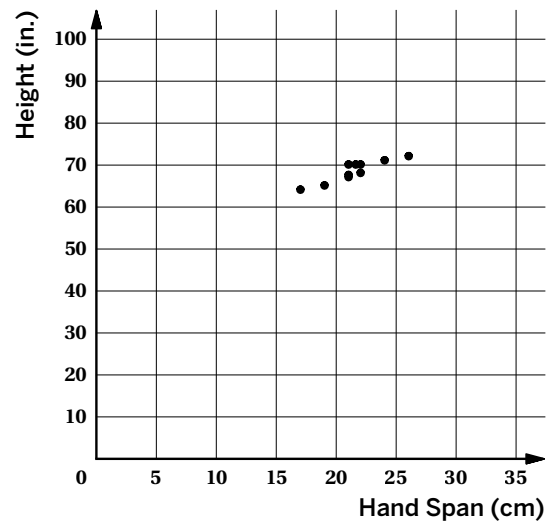


5. Here is a scatter plot.

- a Determine the type(s) of association and indicate whether there is any clustering.
- b Describe the association and clustering, if any, in context.



6. Noah creates the scatter plot shown and says that it has a positive, linear association and clustering. He explains that the association means that, as a hand span increases, the height of a person decreases. Do you agree with Noah? Explain your thinking.

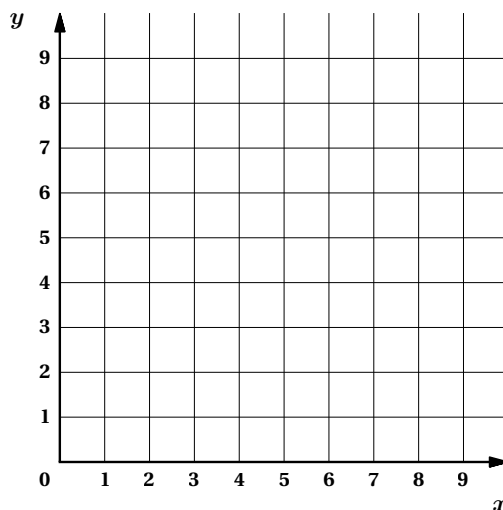


Additional Practice

ACC7.6.09

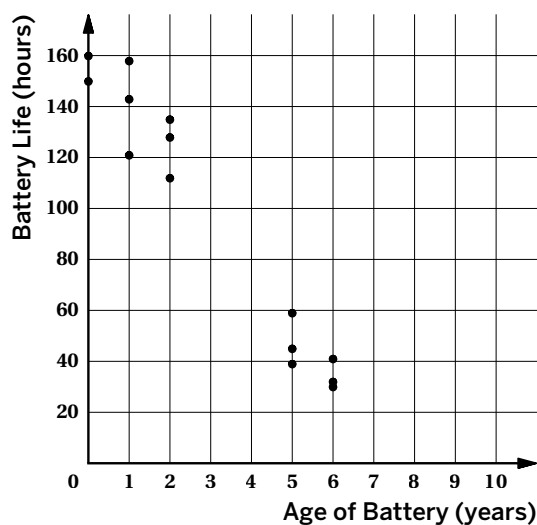
1. Create a scatter plot for this data. Then draw a line on the graph that you think is a good fit for the data.

x	y
2.4	7.6
3.7	8.9
3.1	8
2	4.9
1.5	4



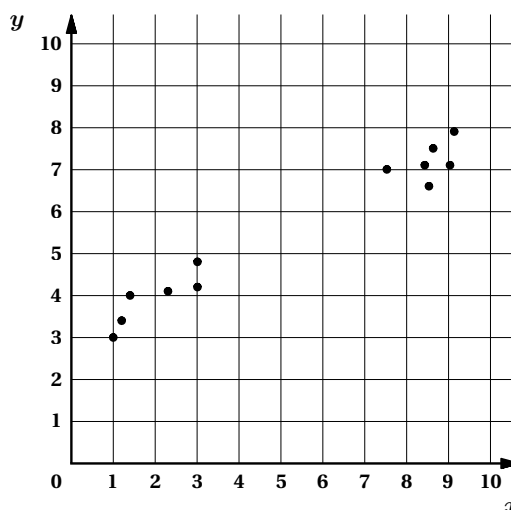
2. Kiran wants to draw a line to model the data shown on the graph. What type of association does there appear to be between the age of a battery and the battery life?

- A. Positive linear association
- B. Negative linear association
- C. No linear association



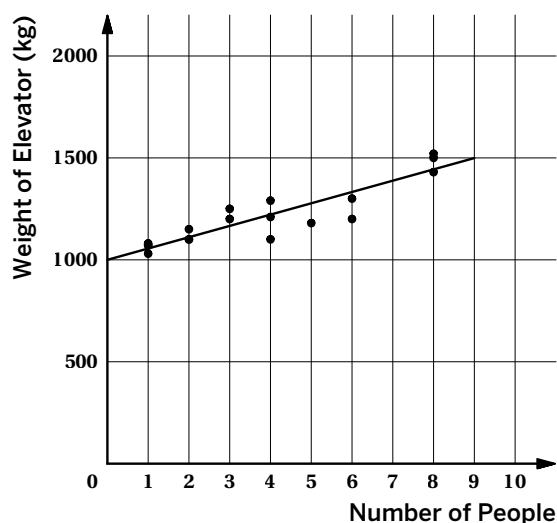
3. For the scatter plot shown, complete the following.

- a Draw a line on the graph that you think is a good fit for the data.
- b A new point is added to the scatter plot where $x = 6$. Use your line to predict the value of y . Add this point to the scatter plot.



4. The scatter plot shows data about the number of people in an elevator and weight in kilograms. This data can be modeled by the equation $y = 53.5x + 1005$.

- a What does the linear model's slope represent in this context?
- b What does the linear model's y -intercept represent in this context?



5. Clare determined the equation for the line of fit of a data set. She noticed that, for a given value of x , her equation predicts a different value of y than is observed in the data. Is her equation incorrect? Explain your thinking.

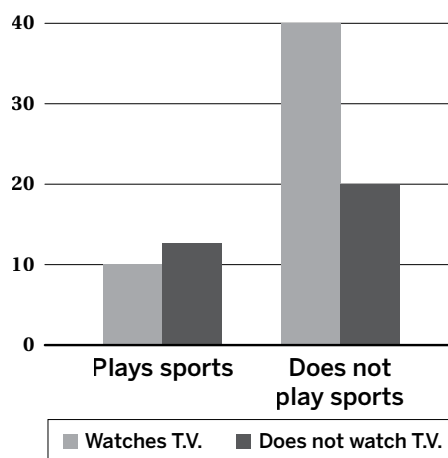
Additional Practice

ACC7.6.10

1. The table shows the results of a survey about news-reading habits among different age groups. What does the number 132 represent in the table?

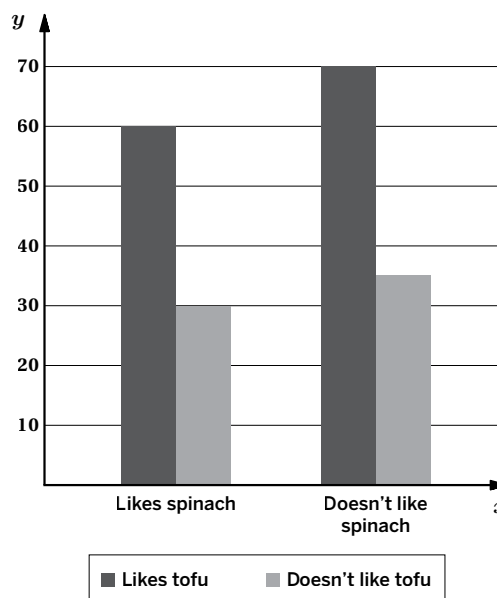
	Internet Media	Print Media	Total
18–25 Year Olds	151	28	179
26–45 Year Olds	132	72	204
Total	283	100	383

- A. 132 people, who are 18–25 years old and read print media
- B. 132 people, who are 26–45 years old and read print media
- C. 132 people, who are 18–25 years old and read internet media
- D. 132 people, who are 26–45 years old and read internet media
2. The bar graph shows data about watching T.V. and playing sports for several students in a middle school. How many students do not play sports and do not watch T.V.?



3. Create table based on the information in the bar graph.

	Likes spinach	Doesn't like spinach	Total
Likes tofu			
Doesn't like tofu			
Total			



4. A restaurant manager records whether people were satisfied with their food and service. Complete the two-way table. Then create a bar graph based on the information in the table.

	Satisfied With the Service	Dissatisfied With the Service	Total
Satisfied With the Food		8	
Dissatisfied With the Food	9		30
Total		29	110



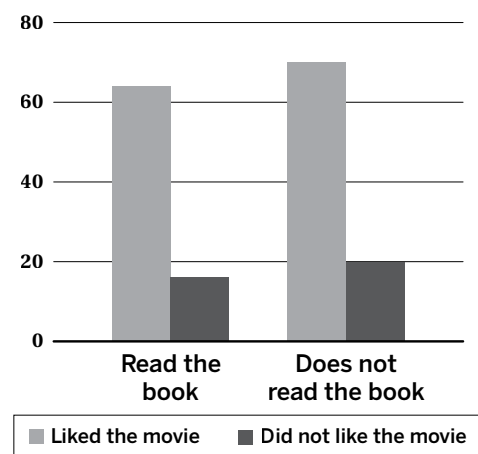
5. A journalist surveys several people about whether they will vote for or against two issues. Complete the two-way table. Then create a bar graph to represent the data.

	For	Against	Total
Issue A	832		997
Issue B		160	
Total	912		1,237



6. A teacher wants to know whether reading a certain book affects liking the movie adaptation of the book for middle school students. Several students' responses were recorded and then graphed using the bar graph shown. What claim might the teacher make based on this data? Explain your thinking.

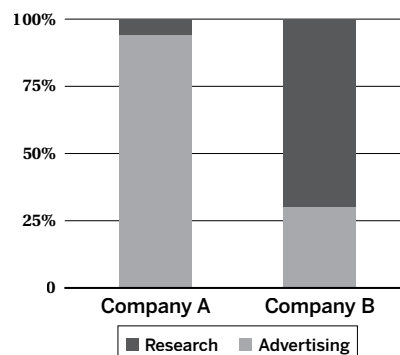
	Liked the Movie	Disliked the Movie	Total
Read the Book	64	16	80
Did Not Read the Book	70	20	90
Total	134	36	170



Additional Practice

ACC7.6.11

- The segmented bar graph displays two companies' yearly budgets, in millions of U.S. dollars. Which of the following describes the data?
 - There is no association; the relative frequency for each company is roughly the same.
 - There is a greater percentage of budget spent on research by Company A.
 - There is a greater percentage of budget spent on advertising by Company A.
 - There is a greater percentage of budget spent on advertising by Company B.



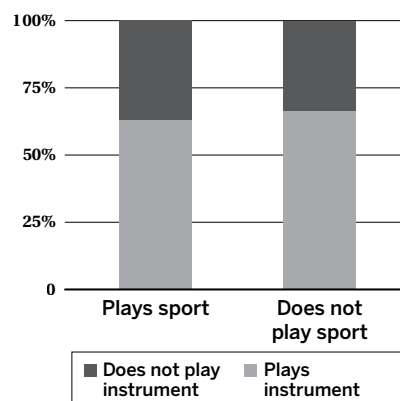
- Complete the table to show the relative frequencies for each column.

	Class A	Class B
Prefers Math	6	8
Prefers Science	2	10
Total	8	18

	Class A	Class B
Prefers Math	75%	
Prefers Science		
Total		

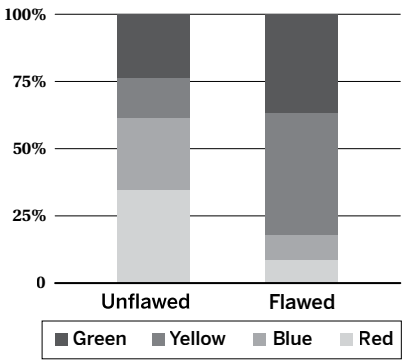
- Students were asked if they play a sport or play a musical instrument. The results are shown in the table and on the segmented bar graph. Is there evidence of an association between playing a sport and playing an instrument? Explain your thinking.

	Plays Instrument	Does not Play Instrument	Total
Plays Sport	12	7	19
Does Not Play Sport	10	5	15
Total	22	12	34



4. The manager of an eraser factory notices some flaws in certain erasers. Each color eraser is made through a different machine. The manager collects data on the number of flawed and unflawed erasers of each color. The results are recorded in the table and are shown on the segmented bar graph. Is there evidence that the flawed erasers are associated with certain colors? Explain your thinking.

	Unflawed	Flawed
Red	285	15
Blue	223	17
Yellow	120	80
Green	195	65



5. A scientist is interested in whether certain plants attract more bees or butterflies. Do these data show an association between bees and butterflies and flower types? Explain your thinking.

	Daisies	Lavender
Bees	17	23
Butterflies	22	28

6. Several students at a middle school were asked whether they preferred drama or band classes. The results are shown in the table.
- Write a question that could be answered by the data in the table.

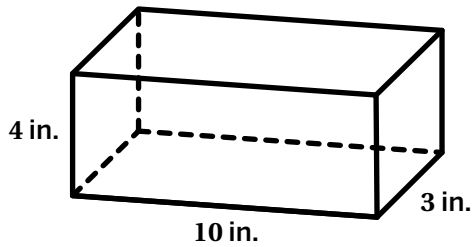
	Drama	Band	Total
7th Grade	72	51	123
8th Grade	85	32	117
Total	157	83	240

Additional Practice

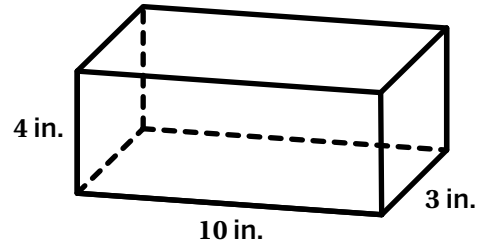
ACC7.7.01

Problems 1–4: Show how to cut a rectangular prism to make each cross section.

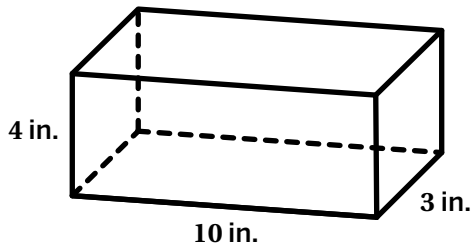
1. A 3-by-4-inch rectangle



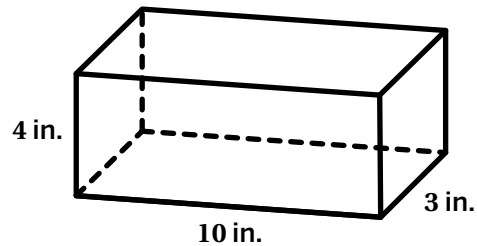
2. A 4-by-10-inch rectangle



3. A different rectangle than Problem 1 or 2

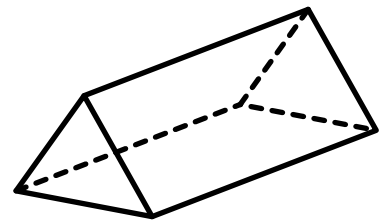


4. A triangle



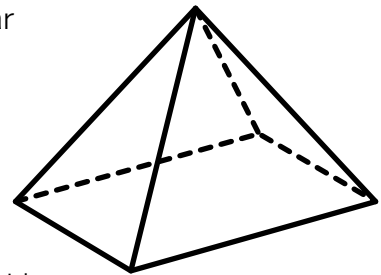
5. Select *all* of the cross sections you can make from the triangular prism shown.

- ☐ A. Triangle
- ☐ B. Square
- ☐ C. Rectangle
- ☐ D. Trapezoid
- ☐ E. Hexagon

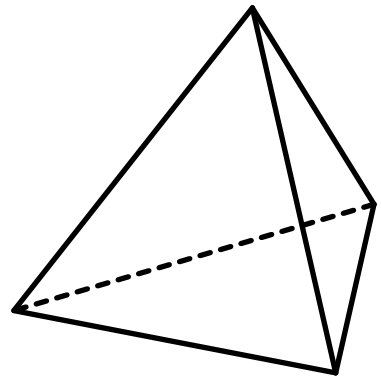


6. Which of the following descriptions could result in a triangular cross section for the rectangular pyramid shown?

- A. Cut the pyramid parallel to its base halfway up.
- B. Cut the pyramid parallel to its base near the top.
- C. Cut the pyramid parallel to its base near the bottom.
- D. Cut the pyramid vertical to its base through the top of the pyramid.



7. Ben says: *No matter which way you cut this triangular pyramid figure, the cross section will be a triangle.* Harper says: *I'm not so sure.* Describe or show a cut that Harper might be thinking of.

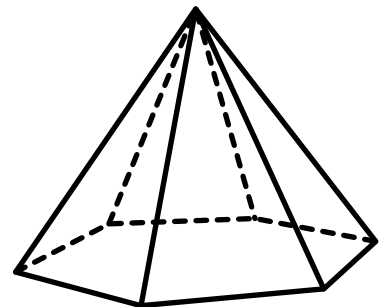


Problems 8–10: The pyramid shown has a hexagonal base. The side lengths of the hexagon are equal. Describe the cross section that will result if the pyramid is cut:

8. Parallel to the base

9. Vertical to the base through the top point of the pyramid

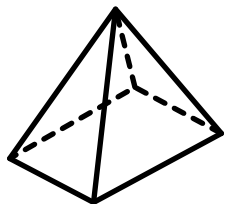
10. Describe another way you could cut the pyramid that would result in a different cross section.



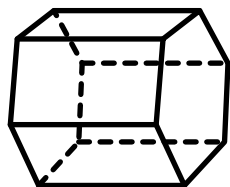
Additional Practice

ACC7.7.02

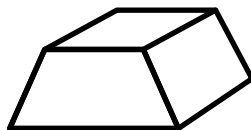
Problems 1–2: Here is a set of 3-D objects.



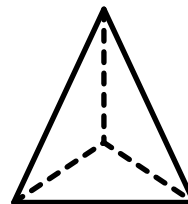
Object A



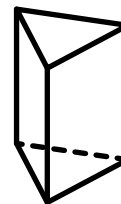
Object B



Object C



Object D

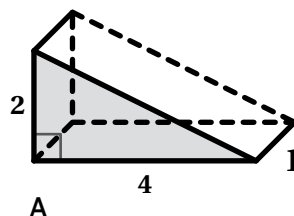


Object E

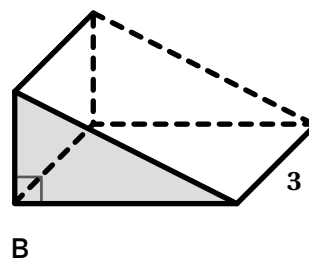
1. Circle all of the prisms.
2. For each prism, shade one of the bases.

Problems 3–5: Here are three prisms with the same base.

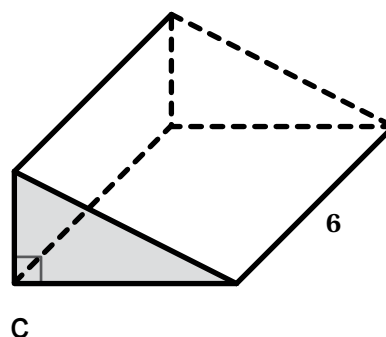
3. Determine the volume of prism *A*.



4. Determine the volume of prism *B*.



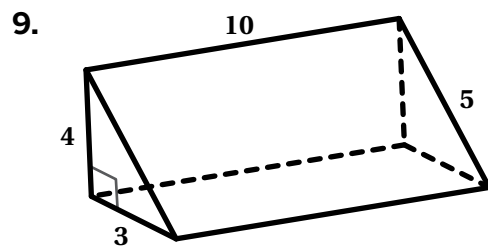
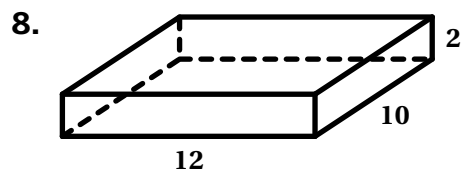
5. Determine the volume of prism *C*.



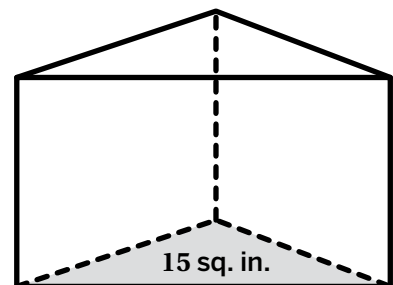
Problems 6–7: The base of a rectangular prism is a square with edges of 4 inches. The height of the prism is 6.6 inches.

6. Draw the prism and label the measurements.
7. Calculate the volume of the prism. Show or explain your thinking.

Problems 8–9: Determine the volume of each prism. Show or explain your thinking.



10. The volume of this prism is between 60 cubic inches and 80 cubic inches. The area of the base is 15 square inches. What is a possible height of the prism? Show or explain your thinking.

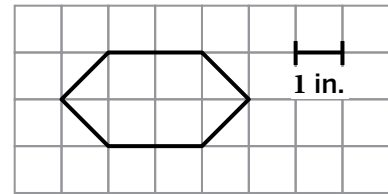


Additional Practice

ACC7.7.03

Problems 1–2: The base of a prism is shown.

1. If the height of the prism is 5 inches, what is the volume of the prism? Show or explain your thinking.



2. If the volume of the prism is 99 cubic inches, what is the height of the prism? Show or explain your thinking.

Problems 3–4: Finnias calculated the volume of the prism but made an error.

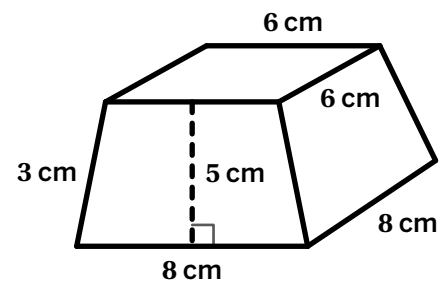
Finnias' Method

$$V = \text{Base} \cdot \text{height}$$

$$V = (8 \cdot 8) \cdot 5$$

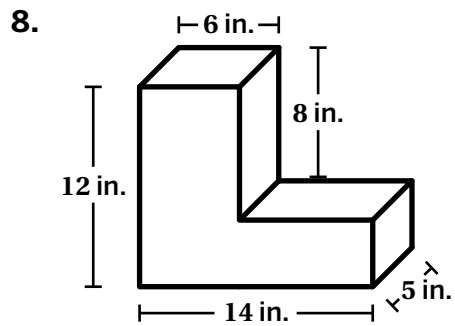
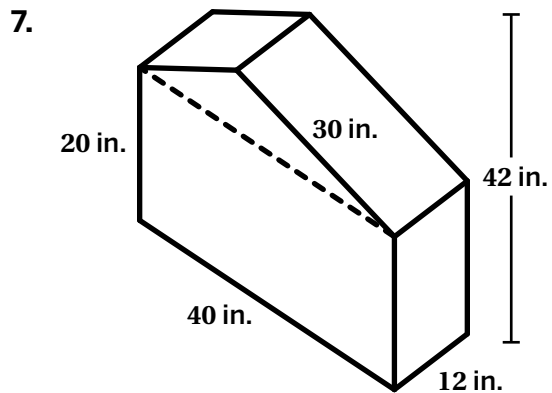
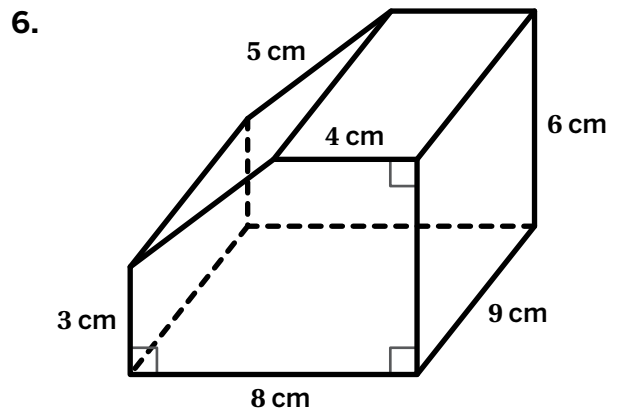
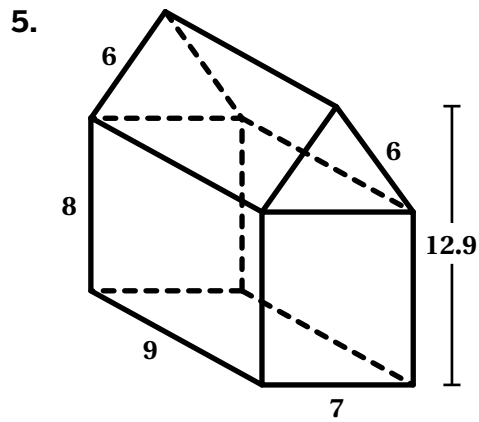
$$V = 64 \cdot 5$$

$$V = 320 \text{ cubic cm}$$



3. Explain why Finnias' method is incorrect.
4. Calculate the volume of the prism.

Problems 5–8: Determine the volume of each prism. Show or explain your thinking.



Additional Practice

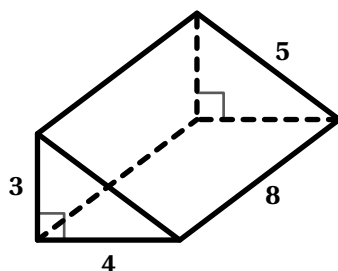
ACC7.7.04

1. Select *all* the situations where knowing the surface area of an object would be useful.

- ☐ A. The amount of cardboard needed to make a box of cereal
- ☐ B. How much wood is needed to build cubed-shaped blocks
- ☐ C. The amount of paint needed to paint a playhouse
- ☐ D. The amount of milk remaining in a jug of milk
- ☐ E. How much brown paper is needed to cover a package
- ☐ F. Advertising space on a highway sign

Problems 2–3: Determine the volume and surface area of each prism. Show your thinking.

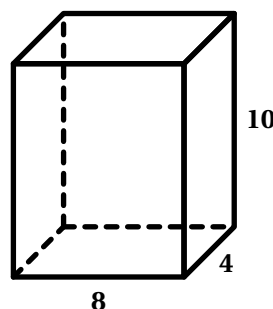
2.



Volume:

Surface Area:

3.



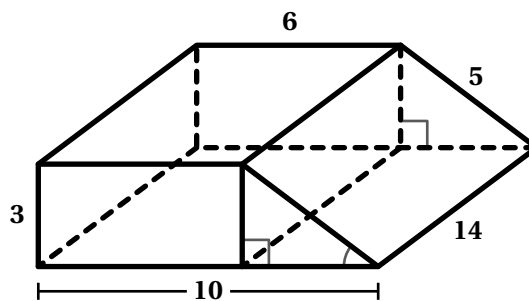
Volume:

Surface Area:

Problems 4–5: Here is a three-dimensional shape.

- 4.** Determine the surface area using two different methods. Show your thinking.

Method #1:



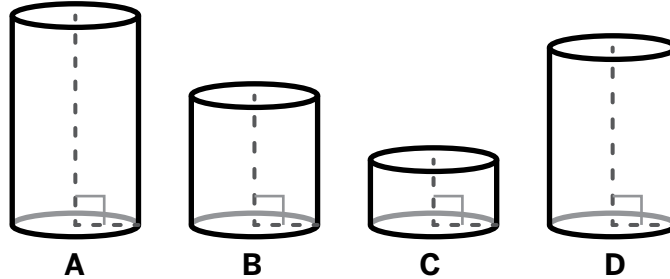
Method #2:

- 5.** Determine the volume. Show your thinking.

Additional Practice

ACC7.7.05

1. Cylinders A, B, C, and D have the same radius but different heights.



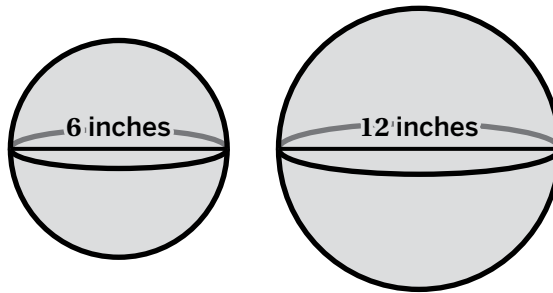
Order the cylinders from *least* volume to *greatest* volume.

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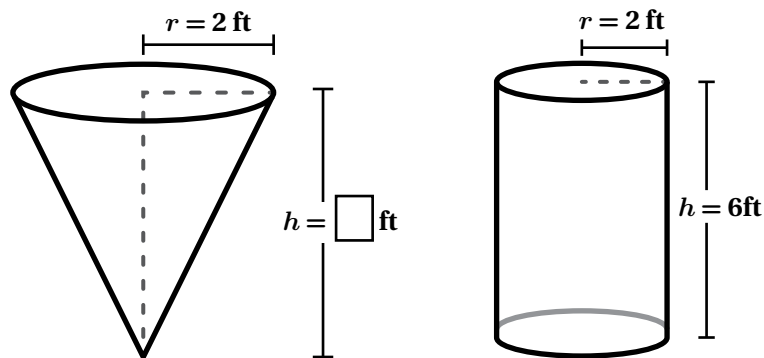
Least

Greatest

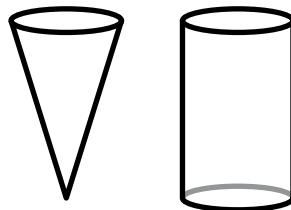
2. The diameter of a balloon measures 6 inches. It has a volume of approximately 113 cubic inches. Kai would like a balloon that has double the diameter. He finds a spherical balloon that measures 12 inches in diameter. What is the approximate volume of this balloon? Explain your thinking.



3. Here is a cylinder and a cone. What does the height of the cone need to be so that the cylinder will have a volume that is three times greater than the cone?



- A. 2 feet
B. 3 feet
C. 6 feet
D. 18 feet
4. Maia is filling a water bottle with a paper cup. The paper cup is in the shape of a cone and has a height of 12 centimeters and a diameter of 6 centimeters. The water bottle is in the shape of a cylinder and has a height of 12 centimeters and a diameter of 6 centimeters. How many paper cups would it take to fill the water bottle? Explain your thinking.

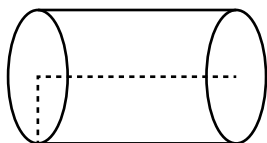


Additional Practice

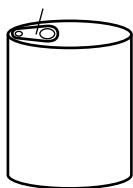
ACC7.7.06

1. For Cylinders A–D, draw the radius and the height. Label each radius with r and each height with h .

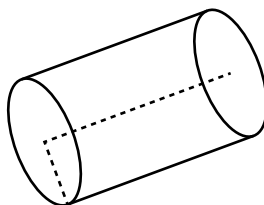
Cylinder A



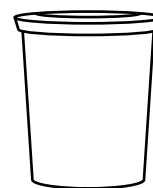
Cylinder B



Cylinder C

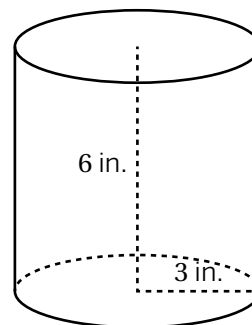


Cylinder D



2. What is the volume of the cylinder shown?

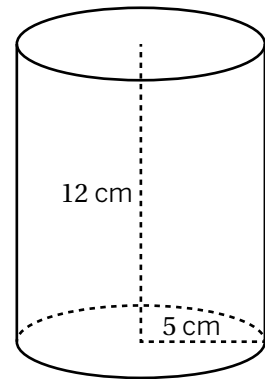
- A. $9\pi \text{ in}^3$
- B. $18\pi \text{ in}^3$
- C. $54\pi \text{ in}^3$
- D. $108\pi \text{ in}^3$



3. **a** Draw a cylinder. Label the radius 4 units and the height 10 units.
- b** Determine the area of the base. Write your response in terms of π .
- c** Determine the volume of the cylinder. Write your response in terms of π .

- 4.** The cylinder shown has a height of 12 cm and a radius of 5 cm.

- a** How many cubic centimeters of fluid can fill this cylinder? Write your response in terms of π .
- b** Give a decimal approximation of your answer to the nearest hundredths place.



- 5.** A preschool teacher stores modeling clay in two types of containers. One container is in the shape of a rectangular prism. The base has side lengths 14 cm and 21 cm. The second container is in the shape of a cylinder with a diameter of 20 cm. Both containers have a height of 8 cm.

- a** Which is greater in area, the rectangular base of the rectangular container or the circular base of the cylinder container? Show your thinking.
- b** Which is greater in volume, the rectangular container or the cylinder container? Show your thinking.

- 6.** Cylinder A has a radius of 9 inches. Cylinder B has the same height as Cylinder A and a radius of one third as long as Cylinder A. What fraction of the volume of Cylinder A is the volume of Cylinder B? Show or explain your thinking.

- 7.** Andre says that if you double the height of a cylinder, it will have the same effect on the volume as doubling the radius of the cylinder. Do you agree with Andre? Show or explain your thinking.

Additional Practice

ACC7.7.07

1. Each row of this table lists information about a specific cylinder. Complete the table.

Diameter (units)	Area of Base (sq. units)	Height (units)	Volume (cu. units)
	9π	3	
4		5	
2			2π
	25π	6	

Problems 2–5: Each soup can is a cylinder with a height of 5 inches.

2. Write an equation that represents the relationship between the volume, V , and the radius, r , for all the soup cans with a height of 5 inches.

3. Complete this table.

r (in.)	4	5	6
V (cubic in.)			

4. The jumbo soup can is also in the shape of a cylinder and has a height of 5 inches, but the radius is 8 inches. What is the volume of this jumbo soup can in cubic inches?

- A. 80π B. 160π
C. 240π D. 320π

5. If the radius of the soup can is doubled, does the volume double?

Yes

No

Maybe

6. A cylinder has a volume of 216π cubic inches and a height represented by h . Complete this table with the volumes of other cylinders that have the same radius but different heights.

Height (in.)	Volume (cu. in.)
$\frac{h}{2}$	
$\frac{h}{3}$	
$2h$	
$3h$	
$4h$	

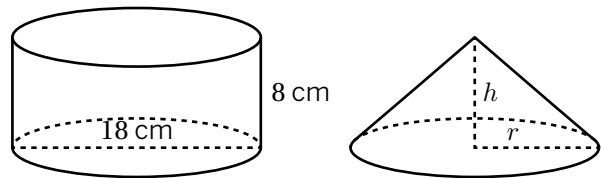
7. Using the volumes from the table, what do you think the volume of a cylinder with a height of $8h$ will be? Explain your thinking.

Additional Practice

ACC7.7.08

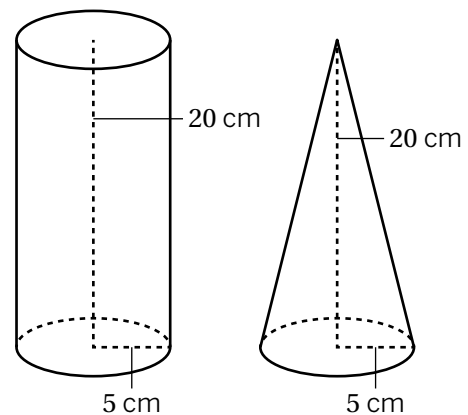
1. The cylinder and cone shown have the same height and the same base area.

- a What is the radius r of the cone?
- b What is the height h of the cone?



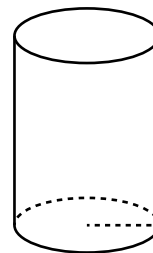
2. Which is a true statement about the volumes of the cylinder and cone?

- A. The volumes are equal.
- B. The volume of the cone is $\frac{1}{3}$ times the volume of the cylinder.
- C. The volume of the cylinder is $\frac{1}{3}$ times the volume of the cone.
- D. The volume of the cone is 3 times the volume of the cylinder.



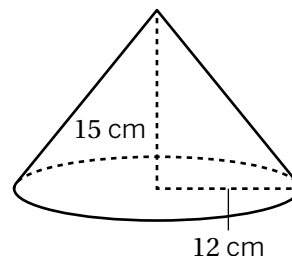
3. The volume of this cylinder is $72\pi \text{ mm}^3$. What is the volume of a cone that has the same base area and the same height?

- A. $24\pi \text{ mm}^3$
- B. $36\pi \text{ mm}^3$
- C. $72\pi \text{ mm}^3$
- D. $216\pi \text{ mm}^3$



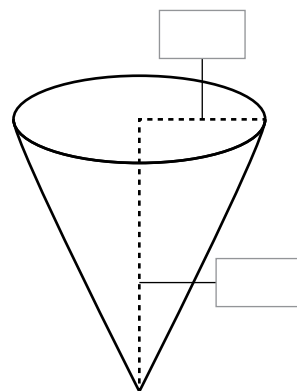
4. What is the volume of the cone shown?

- A. $60\pi \text{ cm}^3$
- B. $180\pi \text{ cm}^3$
- C. $720\pi \text{ cm}^3$
- D. $900\pi \text{ cm}^3$



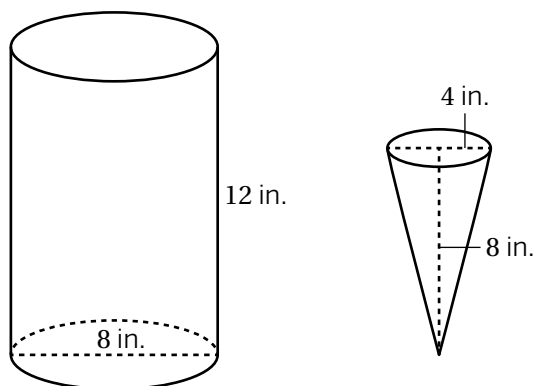
5. A cone-shaped hanging basket is used to grow flowers. The basket has a diameter of 14 inches and a height of 15 inches.

- a Label the height and radius of the hanging basket.
- b If the container is filled completely with potting soil, about how many cubic inches can the container hold? Round to the nearest tenth. Show your thinking.



6. The volume of a cylinder is 96π cubic units. What is the volume of a cone that has the same base area and the same height as the cylinder? Explain your thinking.

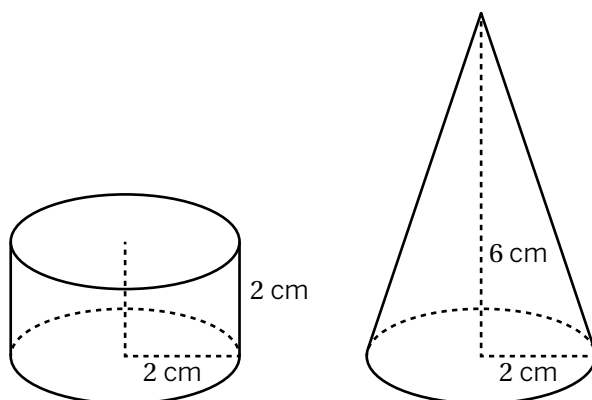
7. At a frozen yogurt shop, a cylinder-shaped container of frozen yogurt has the base diameter and height shown. The shop serves the frozen yogurt in waffle cones that have a base diameter of 4 inches and a height of 8 inches. If each waffle cone is filled completely with frozen yogurt, how many cones can be filled from one container of frozen yogurt? Explain your thinking.



8. Bard claims that the cone has a greater volume than the cylinder. Han argues that the cylinder has the greater volume. Who is correct? Circle one.

Bard Han Both Neither

Explain your thinking.



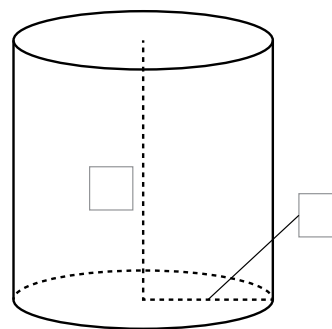
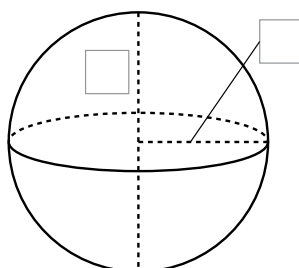
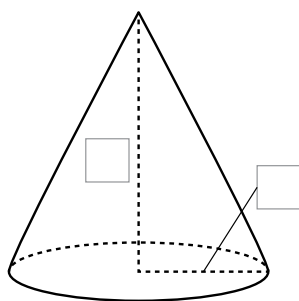
Additional Practice

ACC7.7.10

1. Complete the following table for the dimensions of different spheres.

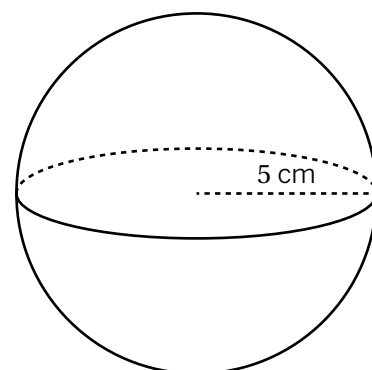
Radius	14 in.		$\frac{500}{3}$ ft		4.09 m	
Diameter		16 cm		$\frac{700}{3}$ yd		4.09 m

2. The cylinder, cone, and sphere all have the same radius and height. The radius of the cone is 7 units. Label the radius and height on each figure, in units.



3. Use the formula $V = \frac{4}{3}\pi r^3$ to determine the volume of the sphere with a radius of 5 cm.

- A. $\frac{125}{3}\pi \text{ cm}^3$
- B. $\frac{500}{3}\pi \text{ cm}^3$
- C. $125\pi \text{ cm}^3$
- D. $500\pi \text{ cm}^3$



4. A sphere has a diameter of 12 inches. What is the exact volume of the sphere? Write your response in terms of π .

5. Which of the following are true statements about the volumes of a sphere, a cone, and a cylinder with the same dimensions? Select *all* that apply.

- ☐ **A.** The sphere has the greatest volume.
- ☐ **B.** The cone's volume is half the sphere's volume.
- ☐ **C.** The cone's volume is half the cylinder's volume.
- ☐ **D.** The sphere's volume is double the cone's volume.
- ☐ **E.** The cylinder's volume is $\frac{3}{2}$ the sphere's volume.
- ☐ **F.** The sphere's volume is $\frac{2}{3}$ the cylinder's volume.

6. Match the description of each sphere to its volume.

- a. Sphere A: radius of 7 cm $\frac{32}{3}\pi \text{ cm}^3$
- b. Sphere B: radius of 2 cm $\frac{256}{3}\pi \text{ cm}^3$
- c. Sphere C: diameter of 8 cm $\frac{1372}{3}\pi \text{ cm}^3$
- d. Sphere D: radius of 9 cm $972\pi \text{ cm}^3$

7. A cube's volume is 216 in³.

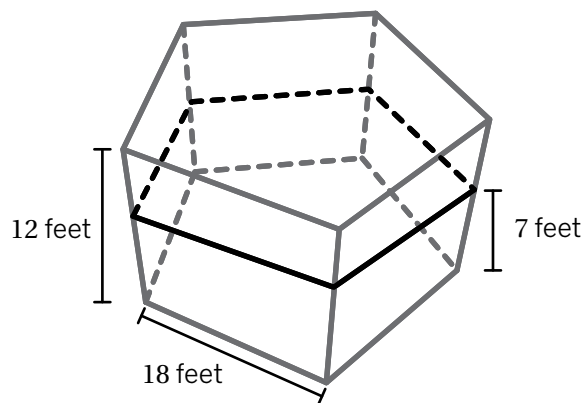
- a** What is the length of its edge?
- b** If a sphere fits snugly inside this cube, what is the volume of the sphere?
Show or explain your thinking.
- c** What percent of the cube is taken up by the sphere? Round to the nearest whole percent.
Show or explain your thinking.

Additional Practice

ACC7.7.11

Problems 1–3: Here is a shed design.

1. 3,899 cubic feet of hay is needed to fill the shed up to 7 feet. What is the area of the shed's base?



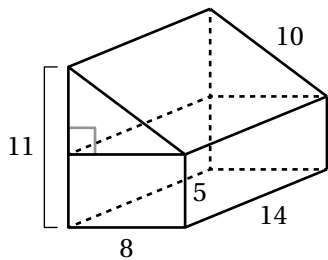
2. If the owner wants to fill the shed up to 12 feet, how much more hay is needed?
3. The owner wants to paint the five sides of the shed (not including the bottom). How many square feet will she need to cover with paint? Explain your thinking.

Problems 4–5: There are two boxes of granola in the shape of rectangular prisms. The first box has a height of 12 inches, a length of 7 inches, and width of 4 inches. The second box has a height of 12 inches, a length of 6.5 inches, and a width of 2.5 inches.

4. What is the difference in volume, in cubic inches, between the two boxes of granola?
5. What is the difference in surface area, in square inches, between the two boxes of granola?

6. Here is a prism

- a** Shade a base of the prism.



- b** Determine the area of the base.
- c** Determine the volume of the prism.
- d** Determine the surface area of the prism.

Additional Practice

ACC7.8.01

1. Match each exponent expression with its expanded expression.

Exponent Expression

Expanded Expression

a. 2^6

..... $3 \cdot 3 \cdot 3 \cdot 3$

b. 6^2

..... $4 \cdot 4 \cdot 4$

c. 4^3

..... $6 \cdot 6$

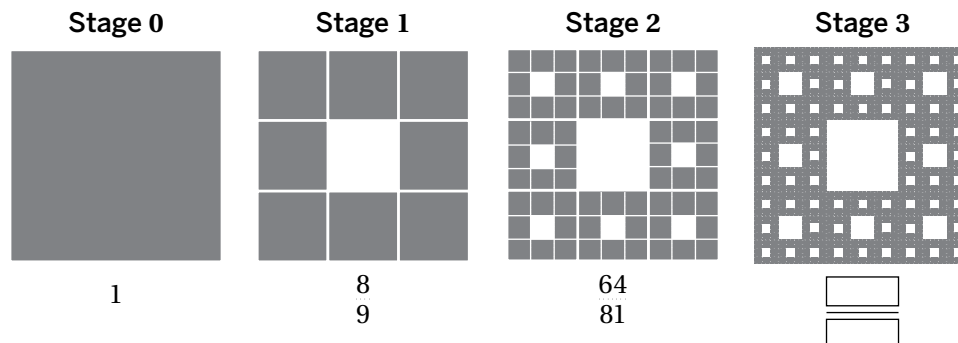
d. 3^4

..... $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$

2. Complete the table.

Exponent Expression	Expanded Expression
10^2	
	$4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$
	$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$
	$\frac{7}{9}$
$\left(1\frac{2}{3}\right)^4$	

3. The pattern shown is called Sierpiński's carpet. Study the pattern. The shaded area of each square in stages 0–2 is written as a fraction of the total area, in square units. Write the fraction that represents the shaded area for stage 3.



Problems 4–6: Write an equivalent expression that uses exponents.

4. $5 \cdot 7 \cdot 5 \cdot 5 \cdot 7 \cdot 5$

5. $4 \cdot 4 + 4 \cdot 4 \cdot 4$

6. 6^4

7. The first three terms of a pattern are shown.

$$\frac{3}{5}, \frac{9}{25}, \frac{27}{125}, \dots$$

- a** What is the 10th term of this pattern? Show or explain your thinking.

- b** Write an expression for the n th term of this pattern. Show or explain your thinking.

Additional Practice

ACC7.8.02

1. For each expression, write an equivalent expression with a single power.

a $7^4 \cdot 7^7$

b $25^8 \cdot 25^3$

c $\left(\frac{2}{3}\right)^9 \cdot \left(\frac{2}{3}\right)^5$

d $(5.4)^{13} \cdot (5.4)^6$

e $(-3) \cdot (-3)^6$

f $6^{15} \cdot 16^7 \cdot 16^3$

2. Select *all* the expressions that are equivalent to 10^{16} .

☐ A. $10^4 \cdot 10^4$

☐ B. $10^2 \cdot 10^8$

☐ C. $10^8 \cdot 10^8$

☐ D. $10^{16} \cdot 10$

☐ E. $10^{15} \cdot 10$

3. Two of the following expressions are equivalent. Circle the expression that is *not* equivalent. Show or explain your thinking.

10^6

$10 + 10 + 10 + 10 + 10 + 10$

$10^3 \cdot 10 \cdot 10 \cdot 10$

4. A large rectangular aquarium is 81 inches long, 81 inches wide, and 9 inches deep. The aquarium is filled to the top with water.

- a Write each measurement of the aquarium as a single power of 9.

Length:

Width:

Depth:

- b Write an expression with exponents to represent the volume, in cubic inches, of the aquarium.

- c How much water does the aquarium hold? Write your response as a single power of 9.

5. Fill in the empty box with a single power of 4 to make each equation true.

a $4^7 \cdot \square = 4^{16}$

b $\square \cdot 4^6 = 4^7$

c $4^{10} \cdot \square \cdot 4 = 4^{18}$

6. Fill in the empty box with a single power of a to make each equation true.

a $a^9 \cdot \square = a^{10}$

b $\square \cdot a^{10} = a^{20}$

c $a^3 \cdot \square \cdot a = a^6$

7. Lin wants to write a multiplication expression that is equivalent to 2^8 . She writes the expression $2^4 \cdot 2^2$. Is her expression correct? Explain your thinking and correct Lin's expression, if necessary.

8. If $a^b \cdot a^b = a^c$ is true, is $\frac{c}{2}$ greater than, less than, or equal to b ? Show or explain your thinking.

Additional Practice

ACC7.8.04

1. Select *all* the expressions that are equivalent to 12^4 .

☐ A. $\frac{12^7}{12^3}$

☐ B. $2 \cdot 6^4$

☐ C. $12^{10} - 12^6$

☐ D. $12 \cdot 12^3$

☐ E. $(12^2)^2$

Problems 2–9: Rewrite each expression as a single power.

2. $3^2 \cdot 4^2$

3. $\frac{7^{12}}{7^8}$

4. $(16^2)^3$

5. $\frac{5^8 \cdot 5^4}{5^9}$

6. $31^2 \cdot 31^{10}$

7. $\frac{9^5}{9}$

8. $(11^4)^5 \cdot 11$

9. $\frac{8^{11}}{4^{11}}$

10. Michael and Amelia were asked to rewrite the expression $\frac{4^{10}}{4^4} \cdot 4$ as a single power.

Michael says the answer is 4^7 . Amelia says the answer is 2^{14} .

Who is correct? Circle one.

Michael Amelia Neither Both

Show or explain your thinking.

11. What is the value of m in the equation $5^m = 5^8 \cdot 5^{10}$.

12. Order the expressions from *least* to *greatest*.

$3^7 \cdot 8^7$	$4^5 \cdot 6^5$	$(24^2)^4$	$24^2 \cdot 24^2$	$24^2 \cdot (24^4)^3$
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Least

Greatest

Additional Practice

ACC7.8.05

1. Rewrite each expression as a single power with a *negative* exponent.

a $\frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10}$

b $\frac{1}{6 \cdot 6 \cdot 6 \cdot 6}$

c $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2}$

d $\frac{1}{y \cdot y \cdot y}$

2. Rewrite each expression as a single power with a *positive* exponent.

a 10^{-2}

b 5^{-5}

c 8^{-7}

d 2^{-8}

3. Rewrite each expression as a single power with a *positive* exponent.

a $10^7 \cdot 10^{-2}$

b $9^{-3} \cdot 9^{-5}$

c $\frac{10^6}{10^8}$

d $\frac{4^2}{4^7}$

4. Select *all* the expressions that are equivalent to $\frac{1}{1000}$.

☐ A. $(-10)^3$

☐ B. 10^{-3}

☐ C. 1000^{-1}

☐ D. 100^{-2}

☐ E. -1000

5. Select *all* the expressions that are equivalent to 10^{-5} .

☐ A. $\frac{10^5}{10^{10}}$

☐ B. -50

☐ C. $10^{-2} + 10^{-3}$

☐ D. $10^{-5} \cdot 10$

☐ E. $\frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10} \cdot \frac{1}{10}$

6. Fill in the empty box with a single power of 10 to make each equation true.

a $\frac{10^3}{\square} = \frac{1}{10^4}$

b $\frac{\square}{10^4} = \frac{1}{10}$

c $10^{-4} \cdot \square \cdot 10 = \frac{1}{10}$

7. Without evaluating, order the expressions 12^{-3} , 12^2 , and 12^0 from *least* to *greatest*. Explain your thinking.

8. Mai states the missing single power of b in the expression $\frac{b^{-12}}{\square} = \frac{1}{b^{10}}$ is b^2 . Do you agree? Show or explain your thinking.

Additional Practice

ACC7.8.07

1. Rewrite each expression three different ways using powers of 10.

a 576,000

.....

b 9,510,000

.....

2. Rewrite each value as a combination of powers of 10.

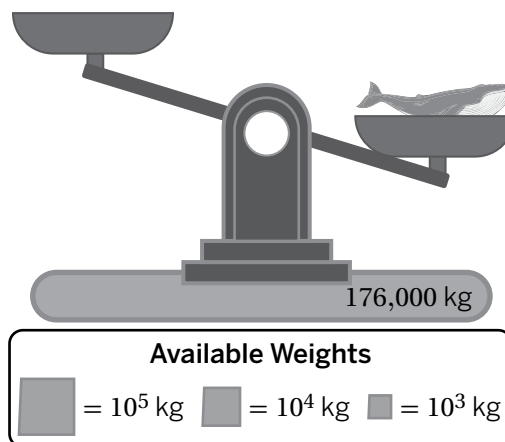
a One light-year is about 6 trillion miles.

b On average, the human brain has 86 billion neurons.

Problems 3–4: Three students tried to balance the scale using weights measuring 10^5 kilograms, 10^4 kilograms, and 10^3 kilograms. They each wrote the weight of the whale as:

- **Expression A:** $176 \cdot 10^2$ kilograms
- **Expression B:** $17.6 \cdot 10^4$ kilograms
- **Expression C:** $1 \cdot 10^5 + 7 \cdot 10^4 + 6 \cdot 10^3$ kilograms

3. Which of these expressions are accurate?



4. Write an expression to represent a different combination of available weights that will balance the scale.

5. Which value is less: $2 \cdot 10^4$ or $2 \cdot 10^5$? Show or explain your thinking.

6. Which value is greater: 500,000 or $50 \cdot 10^4$? Show or explain your thinking.

7. Which of the following numbers has the greatest value?

A. $571.3 \cdot 10^2$

B. $5.713 \cdot 10^3$

C. $571.3 \cdot 10^3$

D. $571.3 \cdot 10^4$

8. Order the expressions from *least* to *greatest*.

$2.612 \cdot 10^5$	$26.12 \cdot 10^2$	$2612 \cdot 10^3$	$2612 \cdot 10^4$
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Least

Greatest

Additional Practice

ACC7.8.08

1. Select *all* expressions that are equivalent to $\frac{1}{100}$.

☐ A. $-1 \cdot 10^2$

☐ B. $1 \cdot 10^{-2}$

☐ C. $-1 \cdot 10^2$

☐ D. $10 \cdot 10^{-3}$

☐ E. $-1 \cdot 10^{-3}$

2. Order the expressions from *least* to *greatest*.

$-4 \cdot 10^0$

$-5 \cdot 10^{-2}$

$4 \cdot 10^2$

$5 \cdot 10^{-2}$

$-4 \cdot 10^2$

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Least

Greatest

Problems 3–4: Write each sum as a decimal.

3. $7 \cdot 10^{-3} + 3 \cdot 10^{-4} + 2 \cdot 10^{-5}$

4. $5 \cdot 10^{-6} + 1 \cdot 10^{-3} + 4 \cdot 10^{-1}$

Problems 5–6: Write each value as a number times a single power of 10.

5. $\frac{8}{100000}$

6. 0.012

Name: Date: Period:

- 7.** Write $-625,000$ three different ways, using a single power of 10.

.....

- 8.** Write 0.000986 three different ways, using a single power of 10.

.....

- 9.** Which value is greater: 0.091 or $9 \cdot 10^{-3}$. Explain your thinking.

- 10.** For what values of x and y will this statement be true: $5 \cdot 10^{-x} > 5 \cdot 10^{-y}$? Show or explain your thinking.

Additional Practice

ACC7.8.09

1. Decide whether each statement is true or false. If the statement is false, explain your thinking.

a 18.2 billion written in scientific notation is 18.2×10^9 .

b 95 million written in scientific notation is 9.5×10^7 .

c 0.00083 written in scientific notation is 83×10^{-4} .

d 0.0005 written in scientific notation is 5×10^{-4} .

2. Complete the table by filling in the scientific notation or equivalent number.

Number	Scientific Notation
23,400	
0.0035	
	6.4×10^7
	9.7×10^{-4}
3,500	
	6×10^{-2}
0.35	
3.5	
350,000,000,000	
	6×10^{-9}

- 3.** Han and Clare were determining the correct way to write the approximate radius of a planet, 71 million meters, in scientific notation. Han wrote the value as 71×10^6 meters. Clare wrote the value as 7.1×10^7 meters. Who is correct? Explain your thinking.
- 4.** Bard and Elena were determining the correct way to write the approximate diameter of a red blood cell, 0.000007 meters, in scientific notation. Bard wrote the value as 7×10^{-6} meters. Elena wrote the value as 0.7×10^{-5} meters. Who is correct? Explain your thinking.
- 5.** The radius of a certain fish egg is 0.0028 meters. Andre claims that the number written in scientific notation is 2.8×10^3 meters. Andre is incorrect. Without evaluating, how do you know he is incorrect? Explain your thinking.
- 6.** Shawn analyzed a microorganism in science class. Using a microscope, Shawn measured the diameter to be 0.0000035 centimeters. Shawn was asked to write this value in meters and rewrote the value as 3.5×10^{-6} meters. Did Shawn correctly rewrite the value in scientific notation? Explain your thinking.
- 7.** The volume of a drop of a liquid is 0.05 milliliters. Priya was asked to write this value in liters, so she rewrote the value as 5×10^{-6} liters. Did she correctly rewrite the value in scientific notation? Explain your thinking.

Additional Practice

ACC7.8.11

1. Which number is greater? Circle one.

$24 \cdot 10^7$

$5 \cdot 10^7$

About how many times greater is it than the other number?

2. Which number is greater? Circle one.

$3 \cdot 10^8$

$6.214 \cdot 10^8$

About how many times greater is it than the other number?

3. Which number is greater? Circle one.

$8.6 \cdot 10^6$

$41 \cdot 10^5$

About how many times greater is it than the other number?

Problems 4–5: Complete each sentence by writing a number in scientific notation.

4. $12.8 \cdot 10^5$ is about times as large as $2.2 \cdot 10^2$.

5. $10.1 \cdot 10^{12}$ is about $2 \cdot 10^8$ times as large as

- 6.** On planet Zerg, there are two different types of alien species, zings and zangs. One zing has a mass of $5.6 \cdot 10^6$ kilograms and one zang has a mass of $2.1 \cdot 10^2$ kilograms. About how many times less in mass is one zang than one zing? Write your answer in scientific notation.
- 7.** The mass of one Brachiosaurus is estimated to have been $7.9 \cdot 10^4$ kilograms. The mass of one antarctic krill is $4.86 \cdot 10^{-4}$ kilograms. About how many times more massive is one Brachiosaurus than one antarctic krill? Write your answer in scientific notation.
- 8.** The mass of one white-toothed pygmy shrew is $4.86 \cdot 10^{-3}$ kilograms. One of the largest blue whales to ever have lived had a mass of about $1.7 \cdot 10^5$ kilograms. About how many times more massive is the largest blue whale than the mass of one white-toothed pygmy shrew? Write your answer in scientific notation.

Additional Practice

ACC7.8.13

1. Decide whether each statement is true or false. Show or explain your thinking.

a $6 \cdot 10^3 + 2 \cdot 10^4 = 8 \cdot 10^7$

b $8 \cdot 10^{-3} + 4 \cdot 10^{-2} = 4.8 \cdot 10^{-2}$

c $5 \cdot 10^3 - 3.3 \cdot 10^4 = 1.7 \cdot 10^2$

2. Evaluate each expression. Write the result in scientific notation.

a $1.9 \cdot 10^{-5} + 8.9 \cdot 10^{-5}$

b $4.9 \cdot 10^6 - 4.1 \cdot 10^5$

c $3.8 \cdot 10^3 + 6.2 \cdot 10^3$

d $5.3 \cdot 10^{-4} - 5.2 \cdot 10^{-4}$

3. Select *all* the expressions that are equal to the expression $1 \cdot 10^5$.

☐ **A.** $7.9 \cdot 10^4 + 2.1 \cdot 10^4$

☐ **B.** $6.3 \cdot 10^5 - 5.3 \cdot 10^4$

☐ **C.** $8 \cdot 10^4 + 2 \cdot 10^4$

☐ **D.** $3 \cdot 10^5 + 7 \cdot 10^5$

☐ **E.** $2.1 \cdot 10^5 - 1.1 \cdot 10^5$

4. Select *all* the expressions that are equal to the expression $6 \cdot 10^{-3}$.

☐ **A.** $8.8 \cdot 10^{-3} - 8.2 \cdot 10^{-3}$

☐ **B.** $7.7 \cdot 10^{-2} - 7.1 \cdot 10^{-2}$

☐ **C.** $3 \cdot 10^{-3} + 3 \cdot 10^{-3}$

☐ **D.** $4 \cdot 10^{-4} + 2 \cdot 10^{-3}$

☐ **E.** $4.7 \cdot 10^{-2} - 4.1 \cdot 10^{-2}$

5. Priya wants to find $4.3 \cdot 10^4 + 3.5 \cdot 10^5$ and writes $4.3 \cdot 10^4 + 3.5 \cdot 10^5 = 7.8 \cdot 10^5$. Explain Priya's mistake and determine the correct sum.

6. Bard writes $2.8 \cdot 10^3 - 2.7 \cdot 10^3$ in scientific notation as $1 \cdot 10^3$. Is Bard correct? Show or explain your thinking.

Additional Practice

ACC7.8.14

Problems 1–4: In 2023, a group of countries had an approximate population of 742.3 million people. Russia, Turkey, Germany, and the United Kingdom had the greatest populations out of all the countries.

1. What was the total population of all four countries? Write your answer in scientific notation.

Country	Population (people)
Russia	$1.5 \cdot 10^8$
Turkey	$8.5 \cdot 10^7$
Germany	$8.4 \cdot 10^7$
United Kingdom	$6.8 \cdot 10^7$

2. What was the total population of the other countries in the group that are not listed? Write your answer in scientific notation.
3. One of the smaller countries in the group is Liechtenstein, which has a population of about 40,000 people. How many times greater is the population of Germany than the population of Liechtenstein?

Problems 4–6: Scientists measured the weight and length of the following sea creatures:

- Blue whale: 400,000 pounds and 100 feet in length
 - Great white shark: 5,000 pounds and 20 feet in length
 - Orca whale: 12,000 pounds and 25 feet in length
 - Manta ray: 6,000 pounds and 29 feet in length
 - Angler fish: 5 pounds and 3.3 feet in length
 - Clownfish: 0.15 pounds and 0.33 feet in length
 - Sea horse: 1 pound and 0.5 inches in length
 - Krill: 0.0044 pounds and 0.2 feet in length
- 4.** What is the sum of the weights of a blue whale, a great white shark, and an orca whale? Write your answer in scientific notation.
- 5.** How much longer is a manta ray than a clownfish? Write your answer in scientific notation.
- 6.** A blue whale can consume an astonishing amount of krill each day. On average, a blue whale eats around 8,000 pounds of krill daily during feeding seasons. Approximately how many krill does a blue whale consume in one day? Write your answer in scientific notation.

Additional Practice

ACC7.9.01

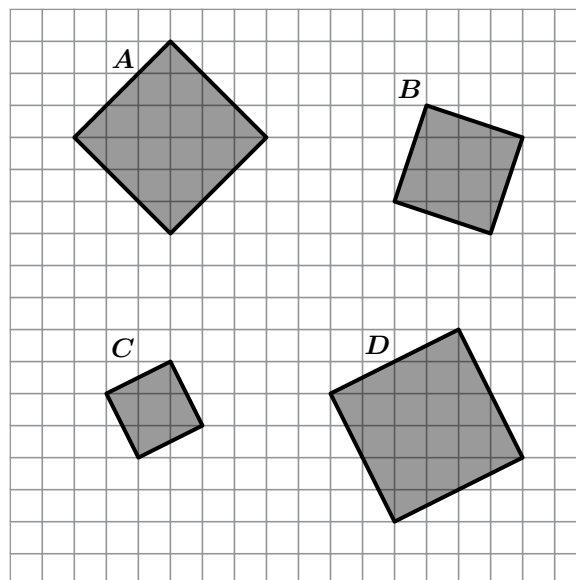
Problems 1–4: Determine the area of each tilted square. Each square grid represents 1 square unit.

1. Square *A*

2. Square *B*

3. Square *C*

4. Square *D*



Problems 5–6: Determine the area of each square given its side length.

5. Side length: 4 centimeters

6. Side length: x units

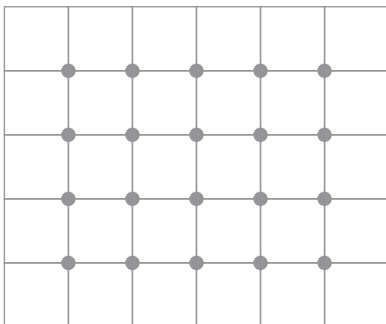
Problems 7–9: Here are the areas of three squares. Determine the side length of each square.

7. Area: 36 square meters

8. Area: $\frac{9}{49}$ square inches

9. Area: w^2 square units

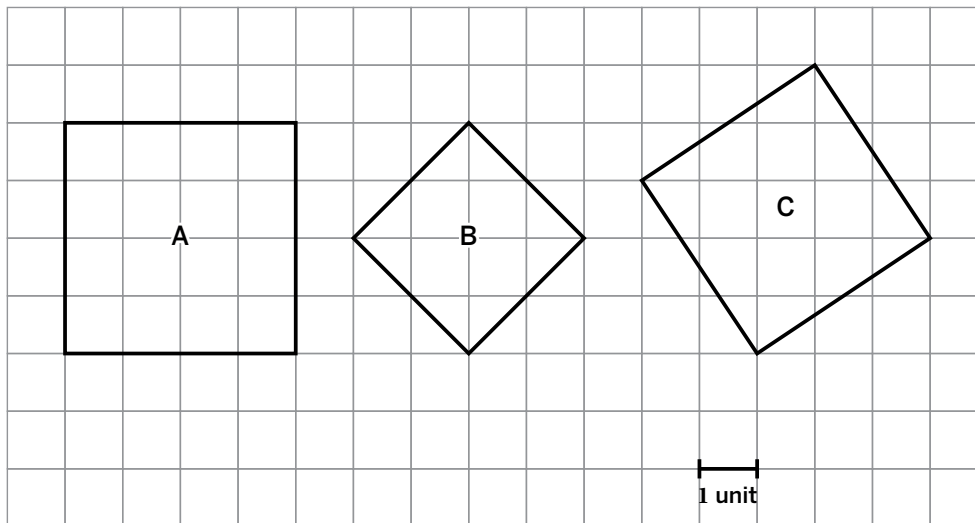
10. Determine the area of the largest tilted square that can be created on the dot grid. Show your thinking.



Additional Practice

ACC7.9.02

1. Determine the area and exact side length of each square.



2. Here are the areas of different squares. Determine the side length for each square.

a Area: 9 square centimeters

b Area: $\frac{1}{4}$ square inches

c Area: 25 square meters

d Area: $\frac{4}{49}$ square feet

3. Here is some information about three squares. Determine which side length matches each square.

- Square A is larger than square B.
- Square B is larger than square C.
- The three squares' side lengths are $\sqrt{10}$, 5.1, and $\sqrt{30}$ units.

Square A: _____ Square B: _____ Square C: _____

4. Square A has an area of 16 square inches. Select *all* the expressions that are equal to the side length of this square (in inches).

☐ A. 2 ☐ B. 4 ☐ C. 8 ☐ D. $\sqrt{4}$ ☐ E. $\sqrt{16}$

5. Here are the side lengths of different squares. Determine the area of each square.

a Side length: 2 centimeters

b Side length: $\frac{1}{3}$ foot

c Side length: $\frac{2}{7}$ inches

d Side length: 0.6 units

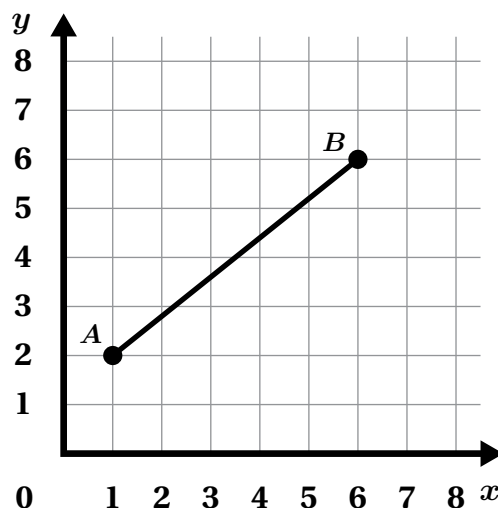
6. Which square has the greatest side length?

- A. Square A with an area of 25 square units
- B. Square B with an area of 64 square units
- C. Square C with a side length of $\sqrt{49}$ units
- D. Square D with a side length of 9 units

Additional Practice

ACC7.9.03

Problems 1–4: Here is a line segment AB . Each grid square represents 1 square unit. Use a ruler, circle, or square if they help with your thinking.



1. Determine the approximate length of AB .
2. Determine the exact length of AB .
3. Which method did you choose to help with your thinking? Circle one.

Ruler
Circle
Square
4. Why did you choose that particular method?

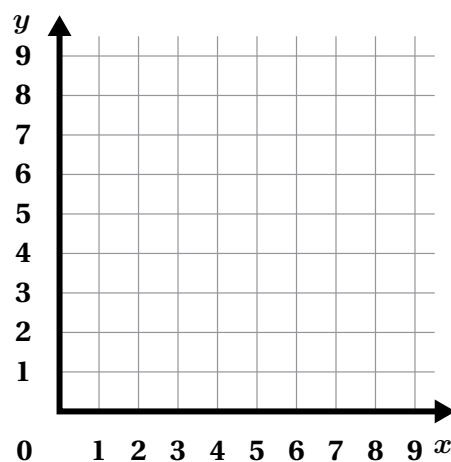
Problems 5–6: Determine the value of each square root.

5. $\sqrt{81}$

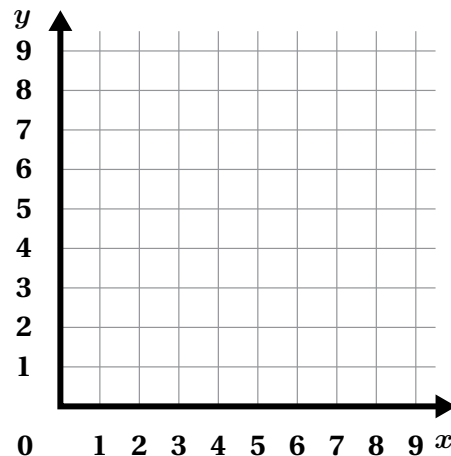
6. $\sqrt{36}$

Problems 7–8: Estimate each square root. Draw a square if it helps with your thinking.

7. $\sqrt{27}$



8. $\sqrt{50}$



Problems 9–10: Determine which two whole numbers each square root is between.

9. $\sqrt{38}$

10. $\sqrt{72}$

11. Here is a list of values ordered from least to greatest. One value is unknown.
Which could be the unknown value?

$\frac{9}{2}, 4.8, ?, \sqrt{27}$

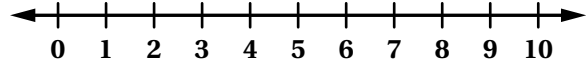
- A. 5.5
- B. $(2.5)^2$
- C. 5.1
- D. $\frac{5}{2}$

Additional Practice

ACC7.9.04

1. Estimate each number and plot it on the number line.

$\sqrt{77}$, 1.5, 6.7, $\sqrt{94}$, $\sqrt{18}$



2. Estimate each square root to the nearest tenth.

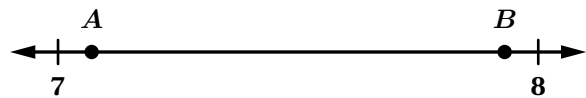
a $\sqrt{13}$

b $\sqrt{28}$

c $\sqrt{78}$

d $\sqrt{85}$

3. Here is a number line with two points plotted between 7 and 8.



- a Write a number using square root notation that could represent point A.

- b Write a number using square root notation that could represent point B.

4. Which statement *best* describes the value of $\sqrt{10}$?

A. The value of $\sqrt{10}$ is between 4.5 and 5.

B. The value of $\sqrt{10}$ is between 4 and 4.5.

C. The value of $\sqrt{10}$ is between 3.5 and 4.

D. The value of $\sqrt{10}$ is between 3 and 3.5.

5. Determine whether each statement is *true* or *false*. Circle your response.

a $\sqrt{14}$ is greater than 3.5. True False

b $\sqrt{125}$ is greater than 11.5. True False

c $\sqrt{22}$ is between 5 and 6. True False

d $\sqrt{77}$ is between 8 and 9. True False

6. Explain how you know that $\sqrt{46}$ is a little less than 7.

7. Select *all* the numbers that are greater than 12 and less than 13.

☐ **A.** $\sqrt{120}$

☐ **B.** $\sqrt{150}$

☐ **C.** $\sqrt{140}$

☐ **D.** $\sqrt{200}$

☐ **E.** $\sqrt{160}$

Additional Practice

ACC7.9.05

1. Write an equivalent expression that doesn't use a cube root symbol.

a $\sqrt[3]{27}$

b $\sqrt[3]{8}$

b $\sqrt[3]{125}$

2. Complete the table. Write an exact value.

x	x^3
1	
$\sqrt[3]{10}$	
	64
	24

3. Order the following from *least* to *greatest*.

$\sqrt[3]{27}$ $\sqrt[3]{120}$ $\sqrt{48}$ $\sqrt{121}$ $\sqrt[3]{\frac{1}{64}}$

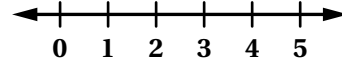
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Least

Greatest

4. Cube A is smaller than cube B . Cube B is smaller than cube C . The edge lengths of the cubes are $\sqrt[3]{62}$ units, $\sqrt[3]{7}$ units, and 8.5 units. Determine the edge lengths of cube A , cube B , and cube C .

5. Here is the equation $x^3 = 33$. Plot the approximate value of x on the number line.



6. Determine whether each statement is *true* or *false*. Circle your response.

- | | | | |
|---|--|------|-------|
| a | $\sqrt[3]{41}$ is less than 3. | True | False |
| b | $\sqrt{4}$ is equal to $\sqrt[3]{8}$. | True | False |
| c | $\sqrt[3]{29}$ is between 3 and 4. | True | False |
| d | $\sqrt[3]{\frac{64}{125}}$ is between 1 and 2. | True | False |

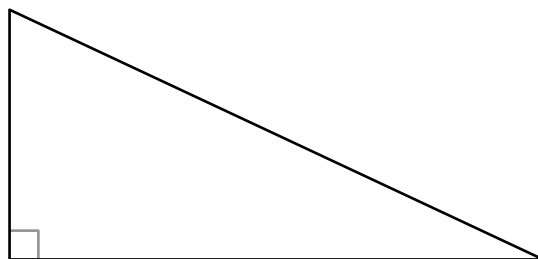
7. Cube E has double the edge length of cube F . What could be the volume of each cube?

8. Cube G has a volume that is double the volume of cube H . What could be the edge lengths of each cube?

Additional Practice

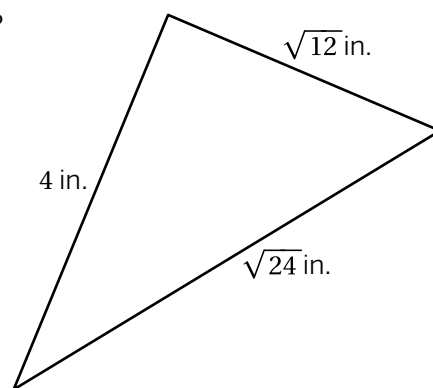
ACC7.9.06

1. Label the legs and hypotenuse of the right triangle.

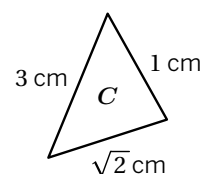
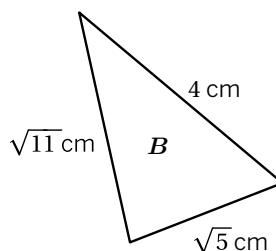
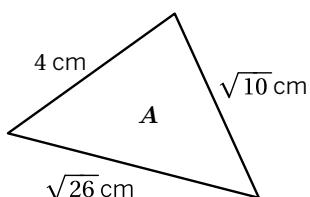


2. Show that the Pythagorean theorem is true for a right triangle with legs of 3 units and 4 units and a hypotenuse of 5 units.

3. Is the Pythagorean theorem true for the triangle shown? Show or explain your thinking.

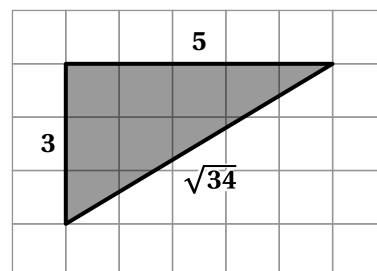


4. For which triangles is $a^2 + b^2 = c^2$ true?



5. Select *all* the equations that represent the relationship between the side lengths of the triangle.

- ☐ A. $(\sqrt{34})^2 = 3^2 + 5^2$
- ☐ B. $5^2 + 3^2 = (\sqrt{34})^2$
- ☐ C. $3^2 + (\sqrt{34})^2 = 5^2$
- ☐ D. $(\sqrt{34})^2 + 5^2 = 3^2$
- ☐ E. $3^2 + 5^2 = (\sqrt{34})^2$



6. Han claims that if $a = \sqrt{5}$, $b = \sqrt{31}$, and $c = 6$, the Pythagorean theorem will be true for this triangle. Is Han correct? Explain your thinking.

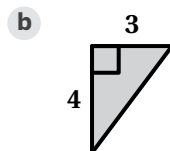
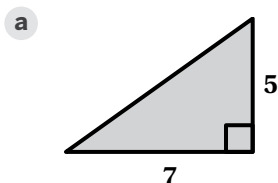
7. Can the Pythagorean theorem ever be true for an isosceles triangle? What about an equilateral triangle? Explain your thinking.

Hint: An isosceles triangle has two congruent sides and an equilateral triangle has three congruent sides.

Additional Practice

ACC7.9.07

1. Determine the length of each unlabeled side.



2. Write an equation that expresses the relationship between the lengths of the three sides of each right triangle.

a $\sqrt{5}, \sqrt{3}, \sqrt{8}$

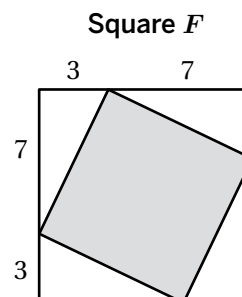
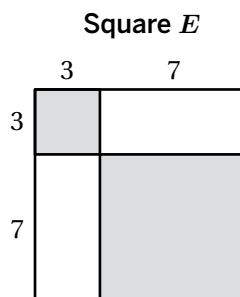
b $1, \sqrt{37}, 6$

c $5, \sqrt{5}, \sqrt{30}$

d $3, \sqrt{2}, \sqrt{7}$

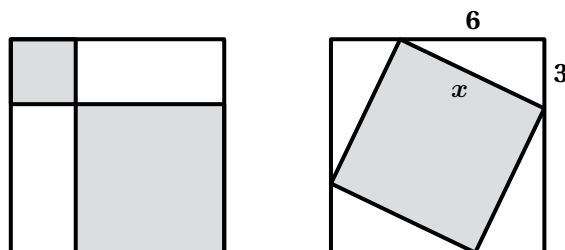
3. Here are squares E and F .

- a** Determine the total area of the shaded region for each square in square units.



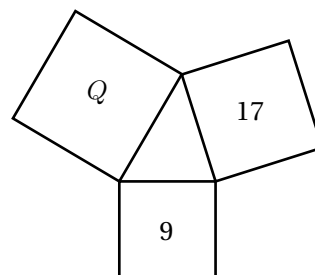
- b** Determine the exact side length of the shaded square inside square F .

4. Determine the exact value of x .

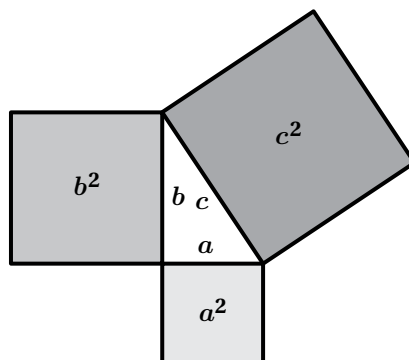


5. The diagram shows an acute triangle and three squares.

Tyler says the area of square Q is 26 square units because $9 + 17 = 26$. Is Tyler correct? Explain your thinking.



6. Determine a set of values for a , b , and c that make $a^2 + b^2 = c^2$ true.



Additional Practice

ACC7.9.08

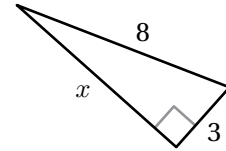
1. Which equation could be used to solve for the unknown side length, x ?

A. $3^2 - 8^2 = x^2$

B. $8^2 + x^2 = 3^2$

C. $8^2 + 3^2 = x^2$

D. $x^2 + 3^2 = 8^2$



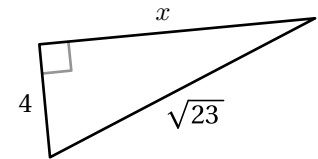
2. Which equation could be used to solve for the unknown side length, x ?

A. $4 + x = \sqrt{23}$

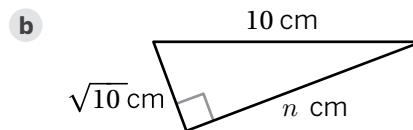
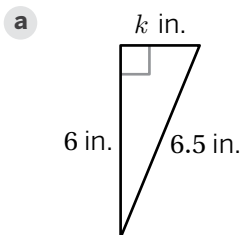
B. $4^2 + x^2 = (\sqrt{23})^2$

C. $4^2 + (\sqrt{23})^2 = x^2$

D. $(\sqrt{23})^2 + x^2 = 4^2$



3. Calculate the exact value of each variable that represents a side length in each right triangle. Show your thinking.

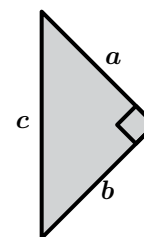


4. Here is a right triangle. Complete each equation to show three relationships among a , b , and c .

a $c^2 = \dots\dots\dots$

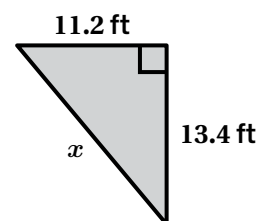
b $a^2 = \dots\dots\dots$

c $b^2 = \dots\dots\dots$

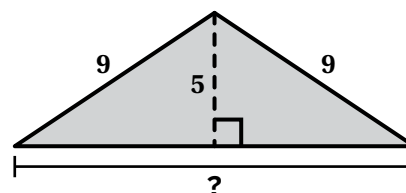


5. This diagram shows a right triangle. Which measurement is closest to the value of x , in feet?

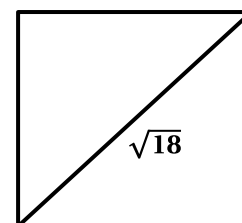
- A. 1.5 feet
- B. 2.2 feet
- C. 17.5 feet
- D. 30.5 feet



6. Calculate the value of the unknown side length.



7. What is the side length of this square?



Additional Practice

ACC7.9.09

1. A triangle has side lengths $\sqrt{11}$, 2, and $\sqrt{15}$ such that $(\sqrt{11})^2 + 2^2 = (\sqrt{15})^2$. Is it a right triangle?

2. The lengths of two sides of a triangle are $\sqrt{7}$ and $\sqrt{32}$. Which side length would make the triangle a right triangle?

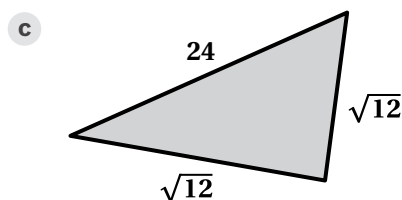
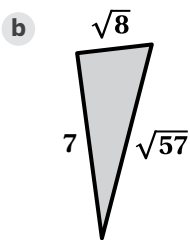
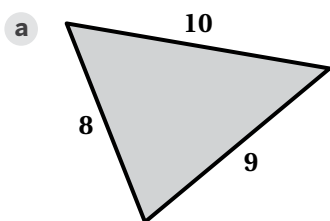
A. $\sqrt{40}$

B. $\sqrt{50}$

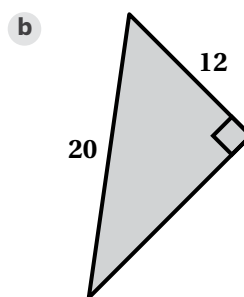
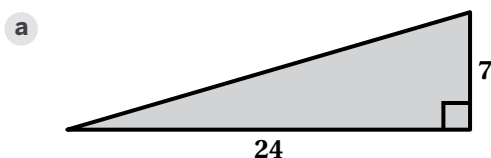
C. 4

D. 5

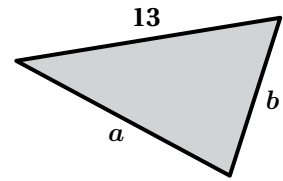
3. Determine whether each triangle is a right triangle.



4. Calculate the value of the unknown side length so that each triangle is a right triangle.



5. The longest side of this triangle has a length of 13 millimeters. What could the lengths of the other two sides be so that this is a right triangle?



6. The side lengths of a triangle are 3, 6, and 10 units. Does the length of the longest side need to *increase*, *decrease*, or *stay the same* to make the triangle a right triangle? Circle one.

Increase

Decrease

Stay the same

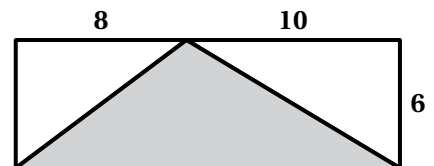
Explain your thinking.

7. Here is a 18-by-6 rectangle divided into triangles. Is the shaded triangle a right triangle? Circle one.

Yes

No

Explain your thinking.



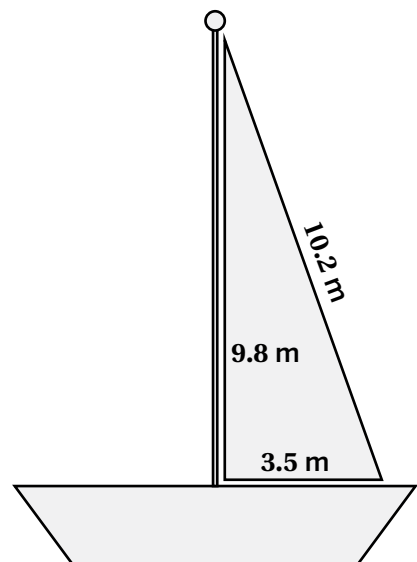
Additional Practice

ACC7.9.10

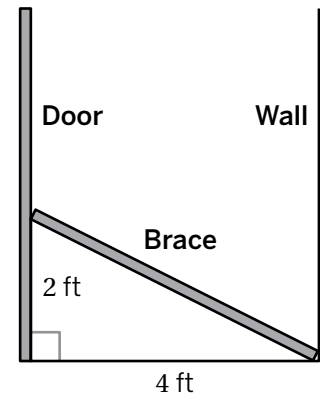
1. Television screens are classified by the length of their diagonal. If a television screen is 22.5 inches tall and 40 inches wide, what is the approximate length of its diagonal in inches? Show your thinking.

2. Lin leaves her house for a jog. She jogs 4 miles directly north, and then 3 miles directly west. If Lin wants to return home, what is the shortest distance she can travel directly back to her house? Show your thinking.

3. Sails come in many shapes and sizes. Is the sail shown a right triangle? Show or explain your thinking.

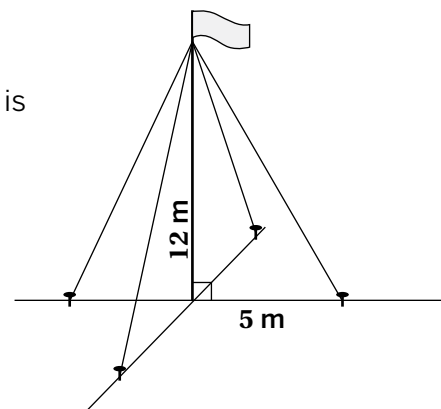


4. A carpenter cuts a length of wood that will brace a door against a wall. The wall is 4 feet away from the door, and she wants the brace to rest on the door, 2 feet above the floor. About how long should she cut the brace, to the nearest tenth of a foot? Show or explain your thinking.



5. Jada is building a marble run track. She wants to create a straight path from the top of one section to the top of a second section. The height of the first section is 8 inches, and the height of the second section is 4 inches. If the distance between the bottom of the two sections is 10 inches, about how long should the connected path be, to the nearest tenth of an inch? Show or explain your thinking.

6. Four cables are used to mount a 12-meter post. Each cable is mounted at the top of the post and extends to the ground, 5 meters from the bottom of the post. What is the total amount of cable needed to mount the post? Show or explain your thinking.



Additional Practice

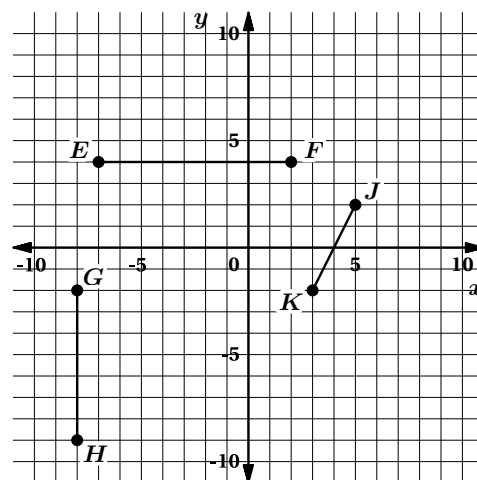
ACC7.9.11

1. Determine the exact length of each line segment.

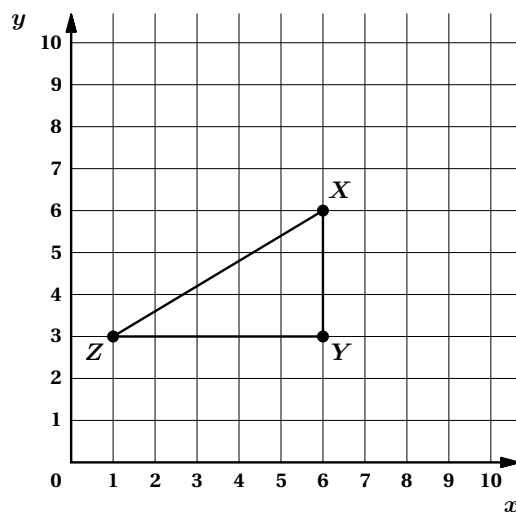
a Line segment EF

b Line segment GH

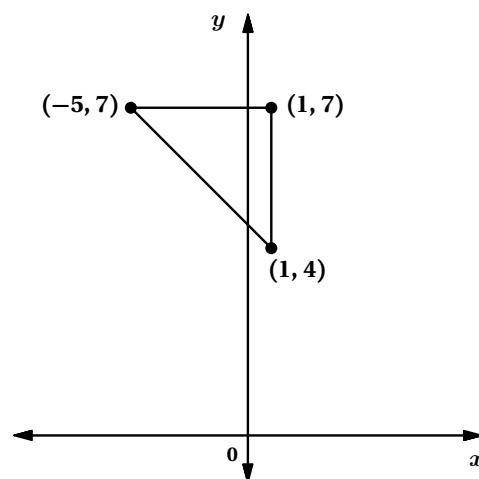
c Line segment JK



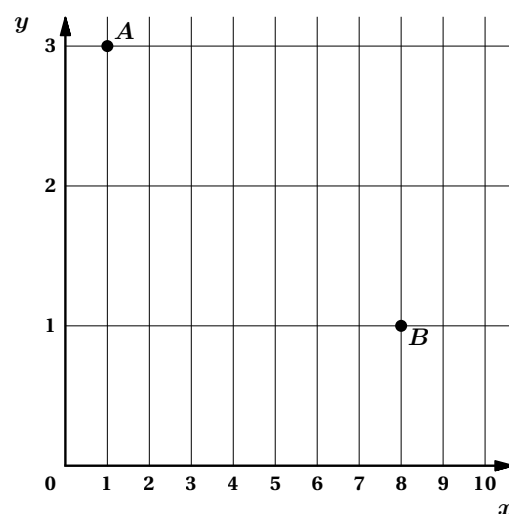
2. Determine the exact length of segment XZ .
Show or explain your thinking.



3. A right triangle is drawn on the coordinate plane, and the coordinates of its vertices are labeled. Label each side of the triangle with its exact length.

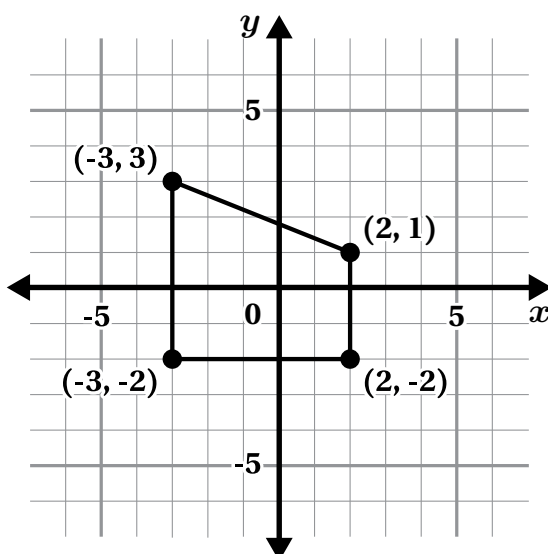


4. Determine the exact distance from point A to point B . Show or explain your thinking.

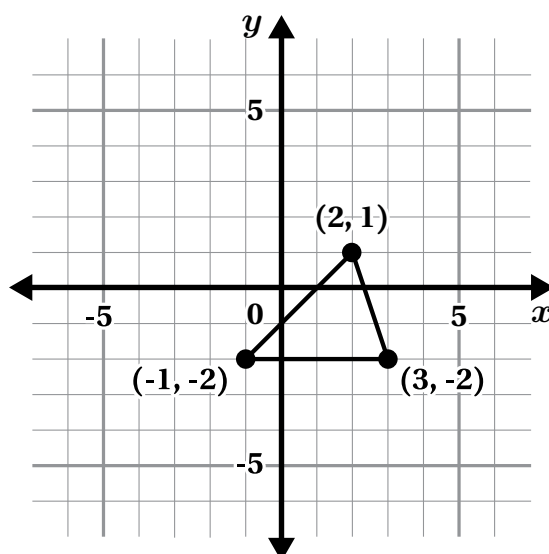


Problems 5–6: Calculate the perimeter of each polygon.

5.



6.



Additional Practice

ACC7.9.12

1. Write each fraction as a decimal.

a $\frac{3}{5}$

b $\frac{23}{100}$

c $\frac{7}{45}$

2. Determine whether the decimal representation of each fraction will repeat or terminate. Circle your response.

a $\frac{1}{20}$ Terminate Repeat

b $\frac{1}{11}$ Terminate Repeat

c $\frac{1}{15}$ Terminate Repeat

3. Which decimal is equivalent to $\frac{4}{27}$?

A. $0.\overline{148}$

B. 0.148

C. $0.14\overline{8}$

D. $\overline{0.148}$

Problems 4–5: Determine whether the decimal representation of each fraction will repeat or terminate. Explain your thinking

4. $\frac{1}{8}$

5. $\frac{1}{18}$

6. Order these numbers from *least* to *greatest*.

$0.\overline{4}$	$0.4\overline{1}$	$0.\overline{41}$	$4.\overline{04}$	0.44
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Least

Greatest

Additional Practice

ACC7.9.13

1. Write each decimal as a fraction.

a 0.4

b $0.\overline{4}$

2. Which fraction is equivalent to $0.\overline{23}$?

A. $\frac{2}{3}$

B. $\frac{23}{99}$

C. $\frac{23}{100}$

D. $\frac{2}{9}$

E. $\frac{3}{2}$

3. Complete the table.

Fraction	Decimal
$\frac{14}{5}$	
$\frac{2}{9}$	
	$-0.\overline{51}$
	$-2.6\overline{7}$
	$0.9\overline{1}$

4. Here are the numbers 0.333 and $0.\overline{3}$.

a How are the numbers alike?

b How are the numbers different?

5. Match each fraction with its decimal representation.

Decimal	Fraction
a. 0.07	 $\frac{7}{100}$
b. $0.0\overline{7}$	 $\frac{5}{9}$
c. $0.\overline{5}$	 $\frac{7}{90}$
d. $0.0\overline{5}$	 $\frac{5}{99}$

6. Determine whether each statement is *always*, *sometimes*, or *never* true. Circle one.

a	A repeating decimal is a rational number.	Always	Sometimes	Never
b	If the digits in the decimal expansion do not repeat and do not terminate, the number is rational.	Always	Sometimes	Never
c	A non-terminating decimal can be written as a fraction.	Always	Sometimes	Never

7. Here are the fractions $\frac{3}{7}$ and $\frac{4}{7}$.

- a Calculate the sum $\frac{3}{7} + \frac{4}{7}$.
- b Write $\frac{3}{7}$ as a decimal.
- c Write $\frac{4}{7}$ as a decimal.
- d Calculate the sum of the decimal representations. What do you notice?

Additional Practice

ACC7.9.14

1. Write three examples of *rational numbers*, including at least one number written as a square root or cube root.
2. Write three examples of *irrational numbers*, including at least one number written as a square root or cube root.
3. Write each number as a fraction of two integers. If it is impossible, write "irrational number."
 - a 0.12
 - b $\sqrt[3]{27}$
 - c $\sqrt{73}$
 - d $-\sqrt{49}$
 - e $\sqrt[3]{11}$
4. Han says that the solution to $x^2 = 50$ is a rational number. Is Han correct or incorrect? Explain your thinking.

5. Sort the numbers based on whether they are rational or irrational.

$\sqrt{9}$	$\sqrt[3]{29}$	0.123	$-\frac{2}{13}$	$\sqrt{8}$	$\sqrt[3]{125}$
Rational			Irrational		

6. Determine whether each statement is *always*, *sometimes*, or *never* true. Circle one.

- | | | | | |
|----------|---|--------|-----------|-------|
| a | Perfect squares are irrational numbers. | Always | Sometimes | Never |
| b | The square root of a number is an irrational number. | Always | Sometimes | Never |
| c | Irrational numbers can be written as fractions. | Always | Sometimes | Never |
| d | The cube root of a perfect cube is a rational number. | Always | Sometimes | Never |

7. Is the length of line segment MN a rational or irrational number? Explain your thinking.

